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Research paper

Mechanical structure design and locomotion control framework for a wheeled hexapod rescue device

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ABSTRACT

This paper presents the mechanical design and hierarchical locomotion control framework of a wheeled hexapod rescue device for disaster-rescue applications. Each wheel-leg adopts a Stewart-structured hybrid mechanism that integrates electric actuation with passive pneumatic transmission, providing multi-degree-of-freedom adjustment for terrain adaptation while supporting payloads with reduced energy demand. A compound power system and an onboard mechanical arm are further integrated to support long-duration operation and rescue manipulation tasks. For wheel-leg motion, a force-position hybrid controller with an admittance outer loop is developed to achieve compliant interaction and accurate trajectory tracking. For whole-body stability, an adaptive model predictive control strategy is established for body-attitude regulation based on rigid-body dynamic analysis and state-dependent reference generation. The mechanical arm is controlled through kinematic modeling and constrained optimization. Experimental results, including energy-efficiency evaluation, terrain-adaptation tests, terrain-traversal trials, teleoperation experiments, and field applications, demonstrate the effectiveness of the proposed system in rescue-oriented locomotion and manipulation tasks.

1. Introduction

Disaster rescue equipment and applications are of particular importance for saving lives and reducing property damage, which have widely attracted sustained attention from researchers and engineers [1–6]. Fixed manual operation tools are initially developed to improve human working ability and efficiency in actual rescue scenes. To enhance the mobility of rescue equipment, the manual mechanisms are mounted on crawler-based chassis [7] or wheeled platforms [8] to constitute mobile apparatuses. Benefiting from a large contact surface touching the ground, adequate grips can be generated on caterpillar treads. Consequently, the tracked system is able to maintain balance stability for traversing extremely tough terrains [9], which is the main reason for a wider range of applications than wheeled vehicles for disaster rescue. However, the heavy caterpillar treads result in problems in high power consumption and increased risk of terrain-induced damage, leading to catastrophic failure in critical locomotion tasks. Recently, there have been significant advances in the field of intelligent robotic systems [10–12]. Different biomimetic robots were investigated to accomplish a series of challenges about autonomous searching and delicate operations (e.g., drilling holes and closing valves) in nuclear accidents or subterranean mishaps [13–15]. Suffering from the shortage of loading capacity, these footed robots are incapable of handling heavy-duty tasks such as transporting supplies, carrying the wounded, and cleaning roadblocks. These requirements indicate that a

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disaster-rescue platform should combine terrain adaptability with sufficient payload capacity and energy efficiency, which motivates the development of wheel-legged systems [16].

Contemporary scholarly efforts in wheel-leg systems predominantly concentrate on innovating kinematic structure design to enhance terrain adaptability [17]. Relevant studies indicate that passive suspensions effectively mitigate the detrimental interactions between wheels and the ground [18,19]. The selection of suspension topology and damping parameters necessitates extensive multi-objective optimization [20,21], revealing inherent limitations in adapting to variable terrain geometries while simultaneously addressing the competing demands of stability, cost efficiency, handling precision, and ride comfort. These challenges consequently catalyze the development of electromechanics integration solutions, particularly semi-active and active wheel-leg architectures with embedded sensing-control capabilities. Such controllable mechanisms are intended to regulate force and position in response to wheel-ground contact, such that the system kinetic energy is dissipated. Extensive studies have been reported on the modeling and controller design of semi-active and active wheel-legs [22–24]. Compared with passive suspended devices, these wheel-legged architectures consume additional energy to counteract gravitational forces acting upon their structural mass and operational payload, reducing endurance time. These developments have further motivated hybrid mobile rescue devices, which are driven by a combination of active and passive components, enhancing terrain adaptability and structural flexibility [25–27]. Advanced wheel-legged platforms employing bionic serial-link legs with wheel-actuator assemblies at the end-effectors have shown strong locomotion capability in unstructured terrain. However, their inherent limitations, particularly the low payload capacity and pronounced fluctuations in body attitude during locomotion [28], render them less suitable for tasks requiring the transport of heavy payloads or the integration of large-scale manipulation tools, further hindering precise operations in disaster rescue scenarios.

Accordingly, a practical wheel-legged rescue device should simultaneously satisfy several requirements: terrain adaptability on uneven ground, sufficient payload capacity for heavy-duty operations, reduced energy demand for sustained deployment, and stable body attitude for manipulation and transport tasks. Existing studies have not fully addressed these requirements within a unified system. To this end, this paper presents a wheeled hexapod rescue device based on a Stewart-structured hybrid wheel-leg mechanism. Each wheel-leg combines electric actuation with passive pneumatic transmission, such that passive support force is introduced to share gravitational and payload loads while retaining multi-degree-of-freedom adjustment capability. In addition, a compound power system is incorporated to support long-duration operation, and a mechanical arm is integrated for rescue manipulation tasks. In terms of the control method, a hierarchical locomotion-control framework is developed to ensure compliant wheel-leg interaction and whole-body attitude stabilization on uneven terrain. For the wheel-leg motion, a force-position hybrid controller based on impedance regulation is employed to ensure compliant interactions and trajectory fidelity. For the body attitude stabilization, an adaptive model predictive control (AMPC) strategy is proposed for real-time attitude regulation, incorporating a state-dependent reference generator derived from centroidal dynamics. The mechanical arm is modeled and controlled through inverse kinematics and constrained optimization, while the locomotion and manipulation processes are coordinated within a distributed framework. The main contributions are summarized as follows:

- A novel wheeled hexapod rescue device is developed, featuring a Stewart-structured hybrid wheel-leg mechanism that integrates electric and pneumatic actuation. The robotic system further incorporates a compound power system and modular control hardware for enhanced adaptability and endurance in disaster environments.
- A hierarchical distributed control framework is proposed, combining an AMPC strategy for body attitude stabilization with force-position hybrid control for each leg, supported by rigorous Lyapunov-based stability analysis.
- The proposed system is validated through comprehensive experiments and field trials, demonstrating effective energy savings, adaptive obstacle-crossing, and reliable task execution in unstructured terrains such as slopes, uneven surfaces, and confined environments.

The remainder of this paper is organized as follows. In Section 2, an overview of the proposed wheeled hexapod rescue device is provided. Section 3 presents the mechanical structure, dynamic modeling, controller design, and stability analysis of the wheel-legs. Section 4 elaborates on the mechanical arm, detailing its kinematic modeling and constrained optimization methods. In Section 5, an AMPC controller is proposed for body-attitude stabilization. Section 6 presents the distributed control framework for coordinated chassis locomotion and manipulation. Experimental results are presented in Section 7 to validate the proposed methods. Finally, Section 8 discusses the main findings and the practical capability boundaries of the proposed system.

Notation: In this paper, we denote $\mathbb{R}$, $\mathbb{N}$, $\mathbb{R}^+$, and $\mathbb{N}^+$ as the sets of real numbers, natural numbers, positive real numbers, and positive natural numbers. For $n, m \in \mathbb{N}^+$, $\mathbb{R}^n$ and $\mathbb{R}^{n\times m}$ represent the sets of n-dimensional real vectors and $n \times m$-dimensional real matrices, respectively. For $a \in \mathbb{R}$, $|a|$ indicates the absolute value of a. For $b \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n\times m}$, we express $\|b\|$ and $\|B\|$ as the Euclidean norm for b and the Frobenius norm for B, respectively. For $s, z \in \mathbb{N}^+$, we denote $[B]_{sz}$, where $s \le n$ and $z \le m$, as the element at the sth row and the zth column in B.

2. Wheeled hexapod rescue device overview

This section presents the physical prototype and main hardware components of the proposed wheeled hexapod rescue device. The electronic circuitry topology of the hybrid-power system and the decoupled locomotion-manipulation workflow adopted in rescue operations are described.

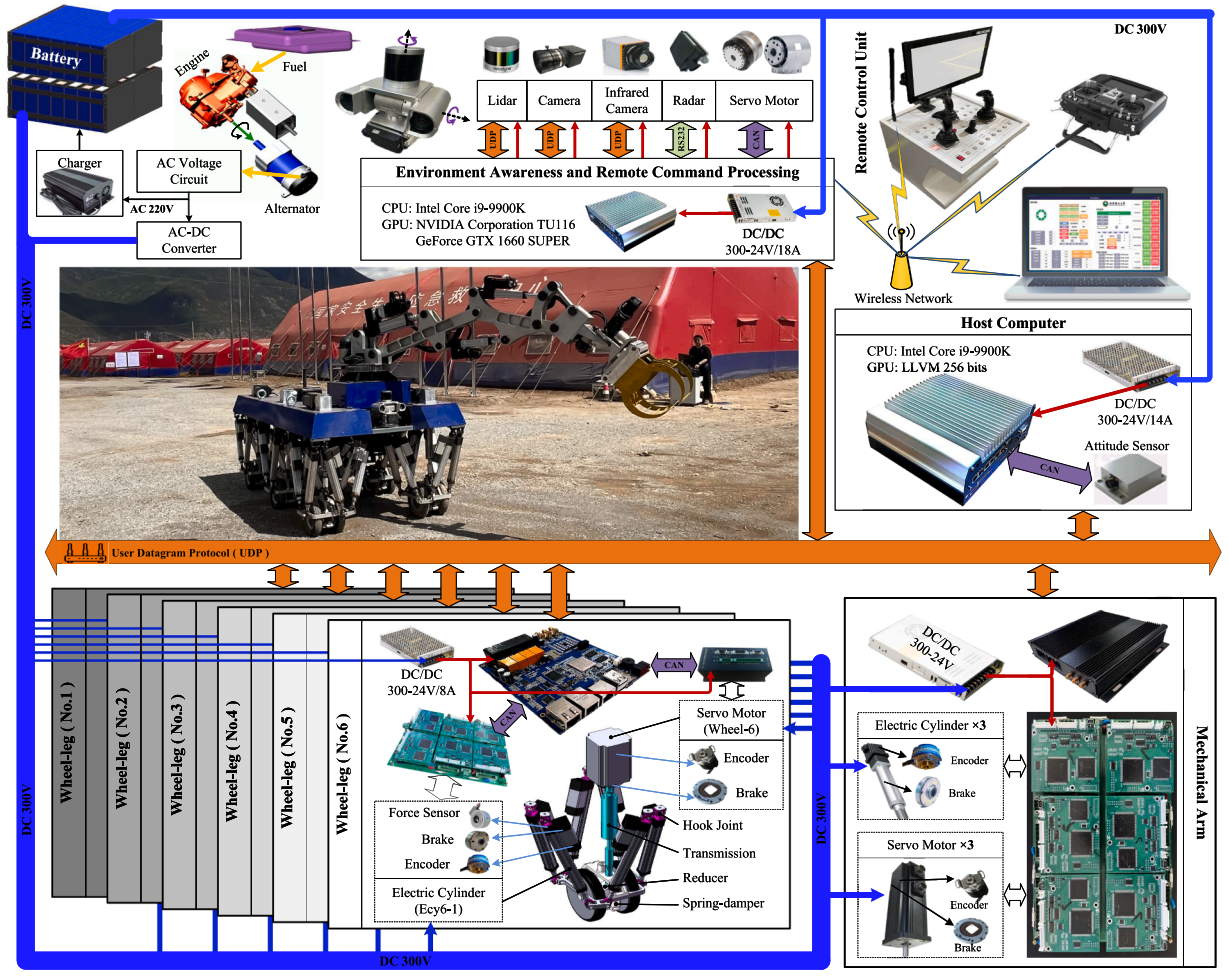


Fig. 1. Physical prototype of the wheeled hexapod robotic system.

2.1. System construction

Theoretical and experimental studies in our previous work verified the feasibility of the Stewart-structured wheel-leg and the corresponding intelligent robotic system, which evolved from quadruped prototypes [29–31] to hexapod configurations [32–34]. To meet the requirements of disaster-rescue tasks, the wheeled hexapod robotic system shown in Fig. 1 was developed.

The physical prototype includes multiple components, enabling excellent locomotion performance, environment information perception, long-term energy supplement, and diverse high-power operations. The host computer fulfills the real-time computation demands by holding an Intel Core i9-9900K CPU with 8 cores, 16 threads, and a base frequency of 3.6 GHz, which communicates with the other components by the user datagram protocol (UDP) and collects sensor feedback information by controller area networks (CAN). The system is equipped with six wheel-legs to provide multiple reconfiguration modes for adapting to complex environments. Equipped with a mechanical arm, the robot handles disaster relief tasks, such as demolition and relocation, with quick-changing engineering accessories. Multiple remote-control units are used to send commands and monitor the system states through a wireless network. Environmental perception and remote-command processing are executed by an Intel Core i9-9900K CPU and a discrete GPU, which improves information-fusion capability.

2.2. Electronic circuitry topology of the hybrid-power system

The wheeled hexapod system is actuated by electricity. To support long-duration operation, the power-management system integrates a lithium battery pack and a generation module. As shown in Fig. 1, the battery package outputs direct current (DC) at 300 V to power the electronic loads, which include the controllers, sensors, and actuators. On the other hand, the combustion engine consumes fuel to drive an alternator, which outputs 220 V alternating current (AC) through a regulation circuit. Through an AC-DC converter, the AC voltage is rectified into 300 V DC, such that the battery and engine can be combined to supply energy to the electronic loads.

Additionally, the battery package can also be charged by the 220 V AC via the charger. As a result, the wheeled hexapod system supports battery-only, alternator-only, and hybrid-power operating modes. In principle, this power architecture extends endurance by enabling alternator-assisted battery charging and increases peak power by allowing the battery pack and the generation module to operate simultaneously.

2.3. Decoupled locomotion-manipulation workflow

The chassis of the wheeled hexapod robotic system comprises the central body and multiple wheel-legs, providing the basis for stable motion over complex terrain. In disaster rescue applications, the Stewart-structured wheel-legs perform adaptive adjustments in response to environmental contacts and body attitude, stabilizing the body and payloads. The mechanical arm is responsible for grasping, transporting, and placing objects. In the operational workflow, the motions of the chassis and the mechanical arm are temporally decoupled and executed independently. During chassis locomotion, the mechanical arm remains stationary in a stowed configuration, and the projection of the combined center of mass stays within the support polygon defined by the ground contact points of the wheel-legs with a prescribed margin, which maintains system stability. This configuration also limits center-of-mass variation during locomotion, keeping the mass and inertia properties effectively constant over the time scale relevant to chassis control. For manipulation tasks, the chassis approaches the target object and stabilizes its attitude. Subsequently, the mechanical arm is commanded to perform grasping, lifting, or relocating, while the chassis remains stationary. This decoupled and independent workflow reduces the coupling effects between the chassis and the mechanical arm, enhancing the locomotion stability and task repeatability of the wheeled hexapod system in field operations.

3. Structure design and control for the wheel-legs

This section presents the mechanical structure and control design of the wheel-legs. The wheel-leg mechanism and its dynamic model are described, followed by the development of a force-position hybrid controller and the corresponding stability analysis.

3.1. Mechanism description and dynamic modeling

The wheel-leg includes six electric cylinders arranged in parallel and a transmission mechanism, which are connected to the robot body by hook joints located at $B_i, P_i \in \mathbb{R}^3$ and denoted by $\overrightarrow{B_iP_i} \in \mathbb{R}^3$ with $i = 0, \dots, 6$. The electric cylinders apply the roller-screw mechanism to supply linear motion driven by motors, which are equipped with force sensors and encoders for state feedback measurement. The servo motor in $B_a \in \mathbb{R}^3$ actuates wheels through transmission $\overrightarrow{B_0P_0}$ and worm-gear reducer at $P_a \in \mathbb{R}^3$. Multiple brakes are also included for emergency safety. The transmission introduces pneumatic passive force to assist the electric cylinders in supporting the body weight, which contributes to high load-bearing capacity and reduced energy consumption. The wheel-leg offers advantages of accurate position decoupling control and high structural stiffness. The number of degrees of freedom (DOFs) $f_{DOF} \in \mathbb{R}$ for the mechanism can be calculated according to the Chebyshev-Grubler-Kutzbach [35] formula, following

$$f_{DOF} = 6 \times (N - m - 1) + \sum_{i=1}^{m} U_i - \lambda, \tag{1}$$

where $N \in \mathbb{R}$ is the sum of links, $m \in \mathbb{R}$ counts the joints, $U_i \in \mathbb{R}$ represents the degrees of the ith joint, and $\lambda \in \mathbb{R}$ denotes the quantity of redundant DOFs. Considering the mechanical details in Fig. 2, the mentioned variables are respectively calculated as $\left[N, m, \sum_{i=1}^{n} U_i\right] = [18, 23, 43]$. The rolling degree for wheels overlaps with coaxial rotation DOF of the leg, which indicates $\lambda = 1$. Then, it has $f_{DOF} = 6 \times (18 - 23 - 1) + 43 - 1 = 6$, such that the wheel-leg maintains 6 DOFs within the operational workspace.

In Fig. 2, coordinate systems Σ_b and Σ_p are set up. The wheel-leg kinematics is modeled as

$$D_i = \| {}^{p}r_b P_i + {}^{p}t_b - B_i \|_2, \tag{2}$$

where $D_i \in \mathbb{R}^+$ with $i \in [1, 6]$ represents $\left|\overrightarrow{B_iP_i}\right|$, ${}^{p}r_b \in \mathbb{R}^{3\times3}$ is the rotation matrix from Σ_p to Σ_b in the order of yaw-pitch-roll, and ${}^{p}t_b \in \mathbb{R}^3$ refers to the translation vector from Σ_p to Σ_b. Since ${}^{p}r_b$ and ${}^{p}t_b$ are functionally related to the generalized position vector $q_p \in \mathbb{R}^6$ for Σ_p in Σ_b, this model mainly concerns the inverse mapping from work-space to joint-space. Then, the wheel-leg tracks desired position trajectory by servo control for electric cylinders with the position references D_i.

The transmission, which links the servo motor and reducer through a hook joint, contains pneumatic components. A hollow piped tank stores compressed gas and supplies it to the cylinder, which adopts a spline between the barrel and piston rod. The wheels gain active torque from servo motor, and a passive force f_a is additionally involved. Based on hydrodynamics theory, we have

$$f_a = \frac{\pi R_c^2 P_c l_t (R_t^2 - R_c^2)}{\left(\left|\overrightarrow{B_0P_0}\right| - l_c\right) R_c^2 + l_t (R_t^2 - R_c^2)}, \tag{3}$$

where $R_c, R_t \in \mathbb{R}$ are the radii of cylinder piston and tank outline respectively, $P_c \in \mathbb{R}$ represents the initial pressure of the pneumatic transmission when the cylinder is fully retracted, $l_t \in \mathbb{R}$ denotes the tank length, and $l_c \in \mathbb{R}$ indicates the cylinder initial length.

In the implementation, the charging pressure P_c is limited to 2.0 MPa by the pneumatic components' maximum allowable working pressure and required safety margins, and the rest parameters in (3) take the following values:

$$R_c = 0.027 \text{ m},\ R_t = 0.032 \text{ m},\ l_t = 0.4 \text{ m},\ \left(\left|\overrightarrow{B_0P_0}\right| - l_c\right) \in [0, 0.6 \text{ m}]. \tag{4}$$

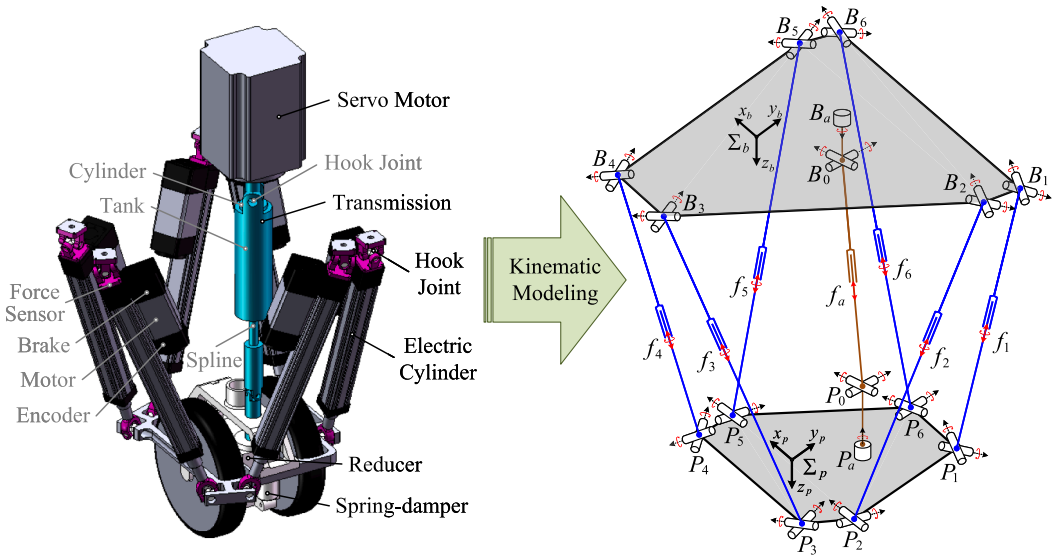


Fig. 2. The mechanical structure and kinematic model of the wheel-leg.

The design principle of P_c is to ensure that the resultant passive force provided by the six transmissions can balance the body mass m_{body} and the mechanical arm mass m_{arm}, which are listed in Table 4. With the gravitational acceleration $g = 9.81\ \text{m/s}^2$, the desired passive force f_a^{d} is given by

$$f_a^{\text{d}} = \frac{(m_{\text{body}} + m_{\text{arm}})g}{6} \approx 1698.8\ (\text{N}). \tag{5}$$

In order to satisfy the basic motion requirements of the robotic system, the wheel-leg extension is typically set to

$$\left(|\overrightarrow{B_0P_0}| - l_c\right) = 0.10\ \text{m}. \tag{6}$$

Substituting (4)–(6) into (3) yields $P_c \approx 1.2$ MPa. Consequently, the initial charging pressure P_c is set to 1.2 MPa.

Based on the Kane method, the wheel-leg dynamics model in joint space is formulated as

$$M_p\ddot{q}_p + C_p\dot{q}_p + G_p = F_e - \tau_e, \tag{7}$$

where $M_p \in \mathbb{R}^{6\times6}$ implies the inertia, $C_p \in \mathbb{R}^{6\times6}$ denotes the Coriolis-centripetal matrix, $G_p \in \mathbb{R}^6$ represents the gravity, $F_e \in \mathbb{R}^6$ represents the joint forces generated by electric cylinders f_i and passive pneumatic f_a, $\tau_e \in \mathbb{R}^6$ is the environmental interaction term, and $q_p \in \mathbb{R}^6$ indicates the position vector for Σ_p in Σ_b. The wheel-leg dynamics satisfy

$$F_e = \begin{bmatrix} \frac{\overrightarrow{B_0P_0}}{|\overrightarrow{B_0P_0}|} f_a \\ \frac{P_0\times\overrightarrow{B_0P_0}}{|\overrightarrow{B_0P_0}|} f_a \end{bmatrix} + \begin{bmatrix} \sum_{i=1}^{6}\left(\frac{\overrightarrow{B_iP_i}}{|\overrightarrow{B_iP_i}|} f_i\right) \\ \sum_{i=1}^{6}\left(\frac{P_i\times\overrightarrow{B_iP_i}}{|\overrightarrow{B_iP_i}|} f_i\right) \end{bmatrix}. \tag{8}$$

For notational brevity, $J_0, f_e \in \mathbb{R}^6$ and $J_w \in \mathbb{R}^{6\times6}$ are written as

$$J_0 = \begin{bmatrix} \frac{\overrightarrow{B_0P_0}}{|\overrightarrow{B_0P_0}|} \\ \frac{P_0\times\overrightarrow{B_0P_0}}{|\overrightarrow{B_0P_0}|} \end{bmatrix},\ f_e = \begin{bmatrix} f_1 & f_2 & f_3 & f_4 & f_5 & f_6 \end{bmatrix}^T,\ J_w = \begin{bmatrix} \frac{\overrightarrow{B_1P_1}}{|\overrightarrow{B_1P_1}|} & \cdots & \frac{\overrightarrow{B_6P_6}}{|\overrightarrow{B_6P_6}|} \\ \frac{P_i\times\overrightarrow{B_iP_i}}{|\overrightarrow{B_iP_i}|} & \cdots & \frac{P_6\times\overrightarrow{B_6P_6}}{|\overrightarrow{B_6P_6}|} \end{bmatrix}. \tag{9}$$

Then, it holds for (8) that

$$F_e = J_0 f_a + J_w f_e. \tag{10}$$

From (7), the system requirement for F_e is determined by supporting the body weight, counteracting external loads, rejecting environmental disturbances, and generating motion. The existence of passive force f_a, arising from the pneumatic components, contributes to joint actuation f_e for fulfilling the demand on F_e, which extends the system endurance time by energy conservation.

The environment interactions are calculated by road spectrum $\lambda_r \in \mathbb{R}^6$ and the spring-damper parameters, which are stiffness $k_s \in \mathbb{R}^{6\times6}$ and damping $b_s \in \mathbb{R}^{6\times6}$. Therefore, τ_e is modeled as

$$\tau_e = k_s(\lambda_r - q_p) + b_s(\dot{\lambda}_r - \dot{q}_p). \tag{11}$$

3.2. Force-position hybrid control

Introduce the system state as $x(t) = q_p \in \mathbb{R}^6$, such that (7) is transformed as

$$M_p\ddot{x} + C_p\dot{x} + G_p = J_0 f_a + J_w f_e - \tau_e. \tag{12}$$

As the electric cylinders track desired position with high stiffness and accuracy, the control signal is defined as the desired position references D_i, following

$$u(t) = [D_1(t), D_2(t), D_3(t), D_4(t), D_5(t), D_6(t)]^T. \tag{13}$$

Ignoring frictional effects, the electric cylinder dynamics indicate

$$M_p\ddot{x} + C_p\dot{x} + G_p = J_0 f_a + J_w\left(f_s + M_e\ddot{u}\right) - \tau_e, \tag{14}$$

where $f_s \in \mathbb{R}^6$ implies the axial forces collected from the force sensors, $M_e \in \mathbb{R}^{6\times6}$ is the mass matrix of the electric cylinders. Then, the state equation is written as

$$\ddot{x} = M_p^{-1} J_w M_e\ddot{u} + f(t, x, \dot{x}), \tag{15}$$

where $f(\dot{x}, x, t)$ is the system function and follows

$$\begin{aligned} f(\dot{x}, x, t) &= M_p^{-1}\left(J_0 f_a + J_w f_s - \tau_e - C_p\dot{x} - G_p\right) \\ &= M_p^{-1}\left[(b_s - C_p)\dot{x} + k_s x - (k_s\lambda_r + b_s\dot{\lambda}_r) + J_0 f_a + J_w f_s - G_p\right]. \end{aligned} \tag{16}$$

For the wheel-leg mechanism, the desired position $x_r(t) \in \mathbb{R}^6$ and force reference $f_r(t) \in \mathbb{R}^6$ are provided as inputs. The position and force tracking errors $e_p, e_f \in \mathbb{R}^6$ are defined as

$$e_p(t) = x(t) - x_r(t) - \int_0^t u_\delta(\tau)\mathrm{d}\tau,\ e_f(t) = f_r(t) - f_s(t), \tag{17}$$

where $u_\delta(t)$ is the displacement adjusting factor for compliant performances corresponding to environmental force interactions.

Addressing force-position hybrid requirements and wheel-leg dynamics in (12), the control signal $u(t)$ is designed for position tracking, following

$$u(t) = \int_0^t M_e^{-1} J_w^{-1} M_p\left[\dot{x}_r(\tau) + u_\delta(\tau) - k_u e_p(\tau) - \int_0^\tau u_{eq}(s)\,\mathrm{d}s\right]\mathrm{d}\tau, \tag{18}$$

where $k_u \in \mathbb{R}^{6\times6}$ is a positive definite diagonal matrix and indicates the controller gain. The equivalent control term $u_{eq} \in \mathbb{R}^6$ holds

$$u_{eq}(t) = f(\dot{x}, x, t) + e_p(t). \tag{19}$$

In order to eliminate any residual constant terms upon differentiation of (18), the integration constants are chosen such that $u(0) = 0$, $\dot{u}(0) = 0$. To further ensure that each wheel-leg actuator achieves accurate position tracking while complying with external force references and exhibiting tunable mechanical compliance, an admittance outer-loop is introduced into the force-position hybrid controller, such that the displacement adjusting factor $u_\delta(t)$ is obtained by processing the force error e_f through a second-order admittance model

$$M_d\ddot{u}_\delta(t) + B_d\dot{u}_\delta(t) + K_\mathrm{d}u_\delta(t) = e_f(t), \tag{20}$$

where $M_d, B_d, K_\mathrm{d} \in \mathbb{R}^{6\times6}$ are positive-definite diagonal matrices representing virtual mass, damping, and stiffness, respectively.

3.3. Stability analysis

For stability analysis, define a Lyapunov function as

$$V(t) = V_1(t) + V_2(t), \tag{21}$$

where V_1 and V_2 are introduced for notational brevity, holding

$$V_1(t) = \frac{1}{2}e_p^T(t)e_p(t) + \frac{1}{2}\dot{e}_p^T(t)\dot{e}_p(t),\ V_2(t) = \frac{1}{2}\dot{u}_\delta^T(t)M_d\dot{u}_\delta(t) + \frac{1}{2}u_\delta^T(t)K_\mathrm{d}u_\delta(t). \tag{22}$$

Then, it is calculated that

$$\dot{V}_1 = e_p^T\dot{e}_p + \dot{e}_p^T\ddot{e}_p = e_p^T\dot{e}_p + \dot{e}_p^T(\ddot{x} - \ddot{x}_r - \dot{u}_\delta). \tag{23}$$

Substituting (18) into (15) yields the closed-loop system dynamics

$$\ddot{x} = \ddot{x}_r + \dot{u}_\delta - k_u\dot{e}_p - e_p, \tag{24}$$

from which $\dot{V}_1(t)$ is rewritten as

$$\dot{V}_1 = -\dot{e}_p^T k_u\dot{e}_p. \tag{25}$$

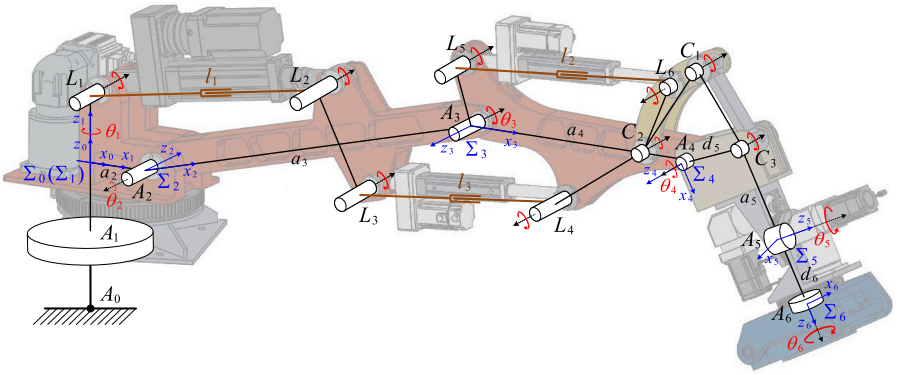


Fig. 3. Kinematic structure and coordinate system for the mechanical arm.

From the admittance model (20), one obtains

$$\ddot{u}_\delta = M_d^{-1}\left(e_f - B_d\dot{u}_\delta - K_\mathrm{d}u_\delta\right). \tag{26}$$

Since $V_2(t)$ is defined by (22), taking the time derivative gives

$$\begin{aligned}\dot{V}_2 &= \dot{u}_\delta^T M_d\ddot{u}_\delta + u_\delta^T K_\mathrm{d}\dot{u}_\delta \\ &= \dot{u}_\delta^T\left(e_f - B_d\dot{u}_\delta - K_\mathrm{d}u_\delta\right) + u_\delta^T K_\mathrm{d}\dot{u}_\delta \\ &= \dot{u}_\delta^T e_f - \dot{u}_\delta^T B_d\dot{u}_\delta.\end{aligned} \tag{27}$$

The combination of (25), (27), and (21) leads to the fact that $\dot{V}(t)$ follows

$$\dot{V} = -\dot{e}_p^T k_u\dot{e}_p + \dot{u}_\delta^T e_f - \dot{u}_\delta^T B_d\dot{u}_\delta. \tag{28}$$

Considering $\dot{u}_\delta^T e_f \le \frac{1}{2}\left(\|\dot{u}_\delta\|^2 + \|e_f\|^2\right)$, we have

$$\begin{aligned}\dot{V} &\le -\lambda_{\min}(k_u)\|\dot{e}_p\|^2 + \frac{1}{2}\left(\|\dot{u}_\delta\|^2 + \|e_f\|^2\right) - \lambda_{\min}(B_d)\|\dot{u}_\delta\|^2 \\ &= -\lambda_{\min}(k_u)\|\dot{e}_p\|^2 + \frac{1}{2}\|e_f\|^2 - \left(\lambda_{\min}(B_d) - \frac{1}{2}\right)\|\dot{u}_\delta\|^2,\end{aligned} \tag{29}$$

where $\lambda_{\min}(k_u)$ and $\lambda_{\min}(B_d)$ denote the minimum eigenvalues of the positive-definite matrices k_u and B_d, respectively. The term $\lambda_{\min}(k_u)$ governs the minimum guaranteed decay rate for the velocity tracking error $\dot{e}_p$, while $\lambda_{\min}(B_d)$ determines the effective damping applied to the compliant displacement dynamics. Provided that k_u and B_d are positive definite and satisfy $\lambda_{\min}(B_d) > \frac{1}{2}$, the derivative of the Lyapunov function $\dot{V}(t)$ remains negative in the presence of a bounded force error e_f, ensuring uniform ultimate boundedness of the system.

Additionally, the admittance model in (20) can be rewritten in state-space form as

$$\begin{bmatrix}\ddot{u}_\delta(t)\\ \dot{u}_\delta(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ -M_d^{-1}K_\mathrm{d} & -M_d^{-1}B_d\end{bmatrix}\begin{bmatrix}\dot{u}_\delta(t)\\ u_\delta(t)\end{bmatrix} + \begin{bmatrix}0\\ M_d^{-1}\end{bmatrix}e_f(t). \tag{30}$$

Due to the positive-definite diagonal M_d, B_d and K_d, the system matrix is Hurwitz. To further guarantee bounded force error responses in practical scenarios, the reference force $f_r(t)$ is designed to be Lipschitz continuous, and the admittance gains are selected to ensure critical or over-damped behavior by choosing the damping gain B_d to satisfy

$$\left[B_d\right]_{ii} \ge 2\sqrt{[M_d]_{ii}[K_\mathrm{d}]_{ii}} \quad \text{with} \quad i = 1,\ldots,6. \tag{31}$$

As a result, the error $e_f(t)$ remains bounded, preserving the stability of the outer-loop compliance controller and the overall closed-loop system. In particular, asymptotic convergence of the trajectory tracking error $e_p(t)$ can be guaranteed when $e_f(t) \to 0$.

4. Kinematic modeling and constrained optimization for the mechanical arm

The mechanical arm is shown in Fig. 3, and its joints are denoted by $A_i, L_i, C_i \in \mathbb{R}^3$. Joints A_1, A_5, and A_6 are driven by motors with transmission gears, whereas joints A_2, A_3, and A_4 are actuated by adjusting the electric-cylinder lengths l_i. Joint A_0 is rigidly attached to the body of the wheel-legged robotic system. Based on the Modified Denavit-Hartenberg (MDH) theory, the kinematics model is established with coordinate systems Σ_i, where i = 0,…,6, and the parameters in Table 1 are calculated by structure dimensions.

Table 1
DH parameters of mechanical arm.

Coordinate System	Σ_1	Σ_2	Σ_3	Σ_4	Σ_5	Σ_6
α/rad	0	$-\frac{\pi}{2}$	0	0	$\frac{\pi}{2}$	$-\frac{\pi}{2}$
a/m	0	a_2	a_3	a_4	a_5	0
θ/rad	θ_1	θ_2	θ_3	θ_4	$\theta_5 - \frac{\pi}{2}$	θ_6
d/m	0	0	0	0	d_5	d_6

Considering the definitions of Σ_i and the parameters in Table 1, the coordinate transformation ${}^{i-1}T_i$ from Σ_{i-1} to Σ_i, which follows the order of $\alpha - a - \theta - d$, is described as

$$ {}^{i-1}T_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & a_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\theta_i)\cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & -\sin(\alpha_{i-1})d_i \\ \sin(\theta_i)\sin(\alpha_{i-1}) & \cos(\theta_i)\sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & \cos(\alpha_{i-1})d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{32} $$

Then, it is calculated that

$$ {}^0T_6 = \prod_{i=1}^{6} {}^{i-1}T_i = \begin{bmatrix} {}^0r_6 & {}^0p_6 \\ 0 & 1 \end{bmatrix}, \tag{33} $$

where ${}^0T_6 \in \mathbb{R}^{4\times4}$ indicates the coordinate transformation from Σ_0 to Σ_6, while the rotation and translation are decoupled into matrices ${}^0r_6 \in \mathbb{R}^{3\times3}$ and ${}^0p_6 \in \mathbb{R}^3$. Define a vector $q_e = [t_x, t_y, t_z, \psi_x, \psi_y, \psi_z]^T$ as the generalized position for Σ_6 in Σ_0, and it is obtained that

$$ {}^0r_6 = R_z R_y R_x, \ {}^0p_6 = [t_x, t_y, t_z]^T \tag{34} $$

where

$$ R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\psi_x) & \sin(\psi_x) \\ 0 & -\sin(\psi_x) & \cos(\psi_x) \end{bmatrix}, R_y = \begin{bmatrix} \cos(\psi_y) & 0 & -\sin(\psi_y) \\ 0 & 1 & 0 \\ \sin(\psi_y) & 0 & \cos(\psi_y) \end{bmatrix}, R_z = \begin{bmatrix} \cos(\psi_z) & \sin(\psi_z) & 0 \\ -\sin(\psi_z) & \cos(\psi_z) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{35} $$

The desired angles θ_i for joints A_i can be determined from the desired position q_e, which is provided by the remote-control unit for rescue operations. Due to the structural properties of serial mechanical arms, the inverse kinematics admits multiple solutions. In practical engineering, the following optimization constraint conditions,

$$ \theta_i \in [\theta_{i,\min}, \theta_{i,\max}], \ \min \sum_{i=1}^{6} \{\theta_i(k) - \theta_i(k-1)\}, \tag{36} $$

arise from the saturation phenomenon for workspace and actuators.

Since joints A_2, A_3, and A_4 are driven by electric cylinders, the length l_i should be further analyzed, such that

$$ l_1 = \left|\overrightarrow{L_1A_2} + \overrightarrow{A_2L_2}\right|, \ l_2 = \left|\overrightarrow{L_5A_3} + \overrightarrow{A_3C_2} + \overrightarrow{C_2L_6}\right|, \ l_3 = \left|\overrightarrow{L_3A_3} + \overrightarrow{A_3L_4}\right|, \tag{37} $$

which are related to the geometric dimensions of the mechanism and joint angles θ_2, θ_3 and θ_4.

5. Model predictive control for attitude stabilization

Maintaining dynamic stability in unstructured terrains requires coordinated wheel-leg actuation for real-time body attitude regulation. To this end, an AMPC framework is developed based on the coupling between wheel-leg contact forces and body attitude dynamics. This section presents the model predictive attitude-stabilization method for the proposed rescue device. The body attitude dynamics are derived from rigid-body analysis, together with the admissible attitude range and conservative slope limit used in controller design, followed by the development of the AMPC controller, including the augmented prediction model, constrained optimization formulation, and adaptive reference generation strategy.

5.1. Body attitude dynamic modeling

As described in Section 2.3, the mechanical arm remains stationary during chassis locomotion, such that the combined center of mass remains approximately constant. In such operating conditions, the robot body equipped with the mechanical arm is modeled as a rectangular rigid body with a uniform mass distribution. As shown in Fig. 4, a body-fixed frame is attached at the center of mass (CoM), in which the roll, pitch, and yaw angles are denoted by φ, γ, and χ, respectively. During wheeled motion, each wheel-leg is perpendicular to the body, thus the wheel-leg forces F_e act normal to the body plane. Consequently, the resultant vertical forces F_i acting at the four corners and two lateral midpoints of the robot body are obtained by subtracting the evenly distributed gravitational

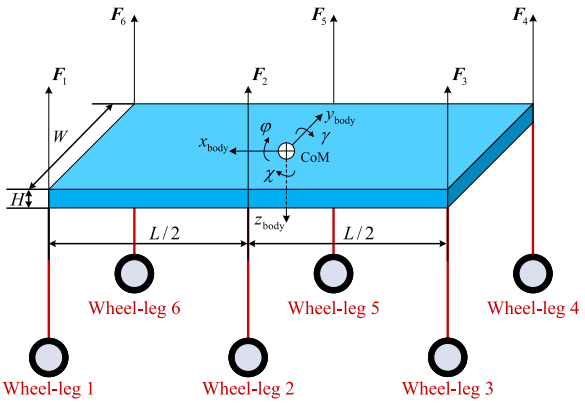


Fig. 4. Rigid body frame and wheel-leg forces.

load:

$$F_i = F_e - \frac{mg}{6}, \quad i \in [1,6], \tag{38}$$

where m denotes the mass of the robot body.

The rigid body rotational dynamics satisfy the Newton–Euler relation

$$M_c = I_c\,\dot{\omega} + \omega \times I_c\,\omega, \tag{39}$$

where $M_c \in \mathbb{R}^3$ is the resultant moment about the CoM, $\omega \in \mathbb{R}^3$ is the angular velocity of the rigid body, and $I_c \in \mathbb{R}^{3\times3}$ is the inertia tensor. When $\varphi = \gamma = 0$, the body axes coincide with the principal inertia axes. Consequently, the initial inertia tensor I_{c_0} in this configuration takes the form of a diagonal matrix, as shown in (40), where the diagonal elements correspond to the moments of inertia along the respective principal axes.

$$I_{c_0} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}. \tag{40}$$

However, when the body attitude changes, the mass distribution of the body relative to the global frame also changes, resulting in the inertia tensor no longer taking a diagonal matrix. During motion over uneven terrain, the yaw angle and yaw rate are kept within small limits to maintain stability, satisfying the small-angle assumption and allowing yaw-induced variations in the inertia tensor to be neglected. Consequently, the model developed in this paper explicitly considers rotations only about the roll and pitch axes, which balances model simplicity and control accuracy. The initial inertia tensor can be transformed by the rotation matrix

$$\begin{aligned} R(\varphi,\gamma) &= R_y(\gamma)R_x(\varphi) \\ &= \begin{bmatrix} \cos(\gamma) & 0 & \sin(\gamma) \\ 0 & 1 & 0 \\ -\sin(\gamma) & 0 & \cos(\gamma) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\varphi) & \cos(\varphi) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\gamma) & 0 & \sin(\gamma) \\ \sin(\varphi)\sin(\gamma) & \cos(\varphi) & -\cos(\gamma)\sin(\varphi) \\ -\cos(\varphi)\sin(\gamma) & \sin(\varphi) & \cos(\varphi)\cos(\gamma) \end{bmatrix}, \end{aligned} \tag{41}$$

yielding the rotated inertia tensor

$$I_c(\varphi,\gamma) = R(\varphi,\gamma)I_{c_0}R(\varphi,\gamma)^T = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix}, \tag{42}$$

where

$$\begin{aligned} I_{11} &= I_{xx}\cos^2(\gamma) + I_{yy}\sin^2(\varphi)\sin^2(\gamma) + I_{zz}\cos^2(\varphi)\sin^2(\gamma), \\ I_{22} &= I_{yy}\cos^2(\varphi) + I_{zz}\sin^2(\varphi), \\ I_{33} &= I_{xx}\sin^2(\gamma) + I_{yy}\sin^2(\varphi)\cos^2(\gamma) + I_{zz}\cos^2(\varphi)\cos^2(\gamma), \\ I_{12} &= (I_{yy} - I_{zz})\sin(\varphi)\cos(\varphi)\sin(\gamma), \\ I_{13} &= (-I_{xx} + I_{yy}\sin^2(\varphi) + I_{zz}\cos^2(\varphi))\sin(\gamma)\cos(\gamma), \\ I_{23} &= (I_{yy} - I_{zz})\sin(\varphi)\cos(\varphi)\cos(\gamma). \end{aligned} \tag{43}$$

To facilitate controller design and real-time computations, the rotated inertia tensor needs to be linearized. Given that the control design aims to suppress body attitude variations induced by ground disturbances in real time, the roll and pitch angles are maintained within small-angle and low angular velocity ranges. To justify this assumption, the maximum allowable roll and pitch angles are derived based on the projection of the CoM relative to the support polygon boundaries, incorporating a specified safety margin. Assuming all feet are in contact and no active attitude regulation, the horizontal residual margins $x(\gamma)$ and $y(\varphi)$ between the CoM projection and the boundaries of the support polygon are formulated as follows:

$$x(\gamma) = \frac{L}{2}\cos(\gamma) - h\sin(\gamma), \quad y(\varphi) = \frac{W}{2}\cos(\varphi) - h\sin(\varphi), \tag{44}$$

where h denotes the CoM height. Substituting $x(\gamma) = d^x_{\text{safe}}$ and $y(\varphi) = d^y_{\text{safe}}$ yields the exact closed-form bounds

$$|\gamma_{\max}| \le \arccos\left(\frac{d^x_{\text{safe}}}{\sqrt{\frac{L^2}{4} + h^2}}\right) - \arctan\left(\frac{2h}{L}\right), \quad |\varphi_{\max}| \le \arccos\left(\frac{d^y_{\text{safe}}}{\sqrt{\frac{W^2}{4} + h^2}}\right) - \arctan\left(\frac{2h}{W}\right), \tag{45}$$

where d^x_{safe} and d^y_{safe} denote the safety margins from the CoM projection to the front–rear and left–right edges of the support polygon, respectively.

In order to ensure sufficient stability margin for controller design, the safety margins d^x_{safe} and d^y_{safe} are set to 1.1 m and 0.6 m, respectively, yielding the selection of $|\gamma_{\max}| = 13°$ and $|\varphi_{\max}| = 6°$. Therefore, the admissible body-attitude range adopted in the following analysis is conservatively defined as a pitch range of $[-13°, 13°]$ and a roll range of $[-6°, 6°]$.

Based on this admissible attitude range, an analytical slope value can be obtained for terrain negotiation with wheel-leg adjustment while maintaining the body attitude within the prescribed range. When only the pitch angle γ is considered, the slope is aligned with the longitudinal axis. The slope angle β is given by

$$\beta = \gamma + \arctan\frac{\Delta l}{d_{\text{wheelbase}}}, \tag{46}$$

where $d_{\text{wheelbase}}$ denotes the wheelbase of the robotic system, and Δl denotes the length difference between the front and the rear wheel-legs. In the extreme configuration, the front and rear wheel-legs saturate the available stroke, with one fully retracted and the other fully extended. Therefore, the maximum usable length difference is

$$\Delta l_{\max} = l_{\max} - l_{\min} = 1.8\ \text{m} - 1.2\ \text{m} = 0.6\ \text{m}, \tag{47}$$

from which the maximum slope angle is derived as

$$\gamma_{\max} + \arctan\frac{\Delta l_{\max}}{d_{\text{wheelbase}}} = 13° + \arctan\frac{0.6\ \text{m}}{1.9\ \text{m}} \approx 30.5°. \tag{48}$$

Similarly, when only the roll angle φ is considered, the slope is aligned with the lateral axis. The maximum slope angle is given by

$$\varphi_{\max} + \arctan\frac{\Delta l_{\max}}{d_{\text{trackwidth}}} = 6° + \arctan\frac{0.6\ \text{m}}{1.2\ \text{m}} \approx 32.6°, \tag{49}$$

where $d_{\text{trackwidth}}$ denotes the track width of the robotic system. To retain safety margins against modeling errors, terrain irregularities, and actuation constraints, the analytical bounds in (48) and (49) indicate that the platform can negotiate slopes of approximately 30° in both longitudinal and lateral directions. These bounds are jointly determined by the admissible attitude limits $\gamma_{\max}$ and $\varphi_{\max}$ and by the geometric ratios $\Delta l_{\max}/d_{\text{wheelbase}}$ and $\Delta l_{\max}/d_{\text{trackwidth}}$.

The maximum approximation errors of the small-angle approximation are given by substituting $\gamma_{\max} = 13° \approx 0.2396$ rad as follows:

$$\begin{aligned} |\sin(\gamma_{\max}) - \gamma_{\max}| &= |\sin(0.2396) - 0.2396| \approx 0.0023, \\ |\cos(\gamma_{\max}) - 1| &= |\cos(0.2396) - 1| \approx 0.0286, \end{aligned} \tag{50}$$

which indicate that the approximation errors remain sufficiently small to be neglected in the subsequent modeling and control design.

As a consequence, the small-angle approximation method can be confidently applied for linearization. Specifically, we approximate $\sin(\varphi) \approx \varphi$, $\cos(\varphi) \approx 1$, $\sin(\gamma) \approx \gamma$, $\cos(\gamma) \approx 1$, while neglecting higher-order small terms (e.g.,φ^2, γ^2, $\varphi\gamma$). The linearized inertia tensor $I_{c,L}(\varphi, \gamma)$ is shown as (51).

$$I_{c,L}(\varphi, \gamma) = \begin{bmatrix} I_{xx} & 0 & (I_{zz} - I_{xx})\gamma \\ 0 & I_{yy} & (I_{zz} - I_{yy})\varphi \\ (I_{zz} - I_{xx})\gamma & (I_{zz} - I_{yy})\varphi & I_{zz} \end{bmatrix}. \tag{51}$$

Let the position vector of each wheel-leg's attachment point to the body be denoted as r_i, then the combined moment is given by

$$M_c = \sum_{j=1}^{6} r_i \times F_i = \frac{1}{2}\begin{bmatrix} W & W & W & -W & -W & -W \\ L & 0 & -L & -L & 0 & L \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix}. \tag{52}$$

According to (39), we have

$$\dot{\omega} = I_{c,L}^{-1}(M_c - \omega \times I_{c,L}\omega). \tag{53}$$

Under small-angle approximations, the rotated inertia tensor remains non-singular, allowing direct inversion. To further simplify model complexity in control law design, higher-order small terms are neglected, resulting in the following approximation

$$\begin{aligned} I_{c,L}^{-1} &= \frac{1}{\det(\mathrm{I_{c,L}})}\begin{bmatrix} I_{yy}I_{zz} - (I_{zz}-I_{yy})^2\varphi^2 & (I_{zz}-I_{xx})(I_{zz}-I_{yy})\varphi\gamma & -I_{yy}(I_{zz}-I_{xx})\gamma \\ (I_{zz}-I_{xx})(I_{zz}-I_{yy})\varphi\gamma & I_{xx}I_{zz} - (I_{zz}-I_{xx})^2\gamma^2 & -I_{xx}(I_{zz}-I_{yy})\varphi \\ -I_{yy}(I_{zz}-I_{xx})\gamma & -I_{xx}(I_{zz}-I_{yy})\varphi & I_{xx}I_{yy} \end{bmatrix} \\ &\approx \begin{bmatrix} \frac{1}{I_{xx}} & 0 & -\frac{I_{zz}-I_{xx}}{I_{xx}I_{zz}}\gamma \\ 0 & \frac{1}{I_{yy}} & -\frac{I_{zz}-I_{yy}}{I_{yy}I_{zz}}\varphi \\ -\frac{I_{zz}-I_{xx}}{I_{xx}I_{zz}}\gamma & -\frac{I_{zz}-I_{yy}}{I_{yy}I_{zz}}\varphi & \frac{1}{I_{zz}} \end{bmatrix}, \end{aligned} \tag{54}$$

$\omega \times I_{c,L}\omega \approx \mathbf{0}$.

Substituting (52) and (54) into (53), the rotational dynamic equations for roll and pitch motions can be rewritten as

$$\begin{bmatrix} \ddot{\varphi} \\ \ddot{\gamma} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \frac{W}{I_{xx}} & \frac{W}{I_{xx}} & \frac{W}{I_{xx}} & -\frac{W}{I_{xx}} & -\frac{W}{I_{xx}} & -\frac{W}{I_{xx}} \\ \frac{L}{I_{yy}} & 0 & -\frac{L}{I_{yy}} & -\frac{L}{I_{yy}} & 0 & \frac{L}{I_{yy}} \end{bmatrix}\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix}. \tag{55}$$

It should be noted that although inertia products arising from roll and pitch rotations were explicitly considered in the derivation, under small-angle approximations and neglecting higher-order infinitesimal terms, contributions from these inertia products vanish. Consequently, the Newton-Euler equations derived under these approximations coincide precisely with the equations obtained by directly employing the initial diagonal form of the inertia tensor. This result confirms that, for small-angle attitude stabilization tasks, inertia product terms exert negligible influence on the linearized system model and may reasonably be disregarded.

Defining the state variable $x = [\varphi, \gamma, \dot{\varphi}, \dot{\gamma}]^T$ and the control input $u = [F_1, F_2, F_3, F_4, F_5, F_6]^T$, the dynamic Eq. (55) can be transformed into a state-space form as follows:

$$\dot{x} = Ax + Bu \tag{56}$$

where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, B = \frac{1}{2}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ W/I_{xx} & W/I_{xx} & W/I_{xx} & -W/I_{xx} & -W/I_{xx} & -W/I_{xx} \\ L/I_{yy} & 0 & L/I_{yy} & -L/I_{yy} & 0 & -L/I_{yy} \end{bmatrix} \tag{57}$$

Using forward differencing, we can rewrite Eq. (56) in the following discrete form

$$x(k+1) = A_k x(k) + B_k u(k), \tag{58}$$

where $A_k = I + TA$, $B_k = TB$, with T being the sampling period.

To ensure stability in attitude control, it is essential to impose precise constraints on the control increment, enabling smooth transitions in control actions. Consequently, the discrete state-space equation above should be further reformulated in terms of the state variable x and the control increment Δu, where $\Delta u(k) = u(k) - u(k-1)$.

Set a new state variable $\xi(k) = [x(k), u(k-1)]^T$, resulting in the following augmented state-space equation

$$\begin{aligned} \xi(k+1) &= \tilde{A}\xi(k) + \tilde{B}\Delta u(k), \\ \eta(k) &= \tilde{C}\xi(k), \end{aligned} \tag{59}$$

where

$$\tilde{A} = \begin{bmatrix} A_k & B_k \\ 0 & I_6 \end{bmatrix}, \tilde{B} = \begin{bmatrix} B_k \\ I_6 \end{bmatrix}, \tilde{C} = \begin{bmatrix} I_4 & 0_{4\times 6} \end{bmatrix}. \tag{60}$$

5.2. AMPC controller design

Define the prediction horizon as p and the control horizon as q ($p \geq q$). Stacking the augmented state and control increment sequences over these horizons yields

$$\begin{aligned} \Xi &= \left[\xi(k+1)^T, \xi(k+2)^T, \ldots, \xi(k+p)^T\right]^T, \\ \Delta U &= \left[\Delta u(k)^T, \ldots, \Delta u(k+q-1)^T\right]^T. \end{aligned} \tag{61}$$

According to the augmented state-space Eq. (59), we have the stacked dynamics

$$\Xi = \mathcal{A}\xi(k) + \mathcal{B}\Delta U, \tag{62}$$

with

$$\mathcal{A} = \begin{bmatrix} \tilde{A} \\ \tilde{A}^2 \\ \vdots \\ \tilde{A}^p \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} \tilde{B} & 0 & 0 & \dots & 0 \\ \tilde{A}\tilde{B} & \tilde{B} & 0 & \dots & 0 \\ \tilde{A}^2\tilde{B} & \tilde{A}\tilde{B} & \tilde{B} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \tilde{A}^{p-1}\tilde{B} & \tilde{A}^{p-2}\tilde{B} & \tilde{A}^{p-3}\tilde{B} & \dots & \tilde{A}^{p-q}\tilde{B} \end{bmatrix}. \tag{63}$$

The objective function of this model predictive controller is chosen in the following quadratic form

$$J(k) = \sum_{i=1}^{p} [x(k+i) - x_{\text{ref}}(k+i)]^T Q [x(k+i) - x_{\text{ref}}(k+i)] + \sum_{i=0}^{q-1} \Delta u(k+i)^T R \Delta u(k+i), \tag{64}$$

where x_{ref} denotes the reference state variable, with $Q \in \mathbb{R}^{4\times4}$ and $R \in \mathbb{R}^{6\times6}$ being the weighting matrices.

Introducing the stacked predicted output

$$X = \left[x(k+1)^T, x(k+2)^T, \dots, x(k+p)^T\right]^T = C\Xi = C\mathcal{A}\xi(k) + C\mathcal{B}\Delta U, \tag{65}$$

with $C = I_p \otimes \tilde{C}$, where $\otimes$ denotes the Kronecker product, the objective function (64) can be rewritten as the following standard quadratic programming (QP) form

$$\min_{\Delta U} \frac{1}{2}\Delta U^T H \Delta U + f^T \Delta U, \tag{66}$$

where

$$\begin{aligned} H &= 2\left[(C\mathcal{B})^T \mathcal{Q} C\mathcal{B} + \mathcal{R}\right], & f &= 2(C\mathcal{A}\xi(k) - X_{ref})^T \mathcal{Q} C\mathcal{B}, \\ \mathcal{Q} &= I_p \otimes Q, & \mathcal{R} &= I_q \otimes R. \end{aligned} \tag{67}$$

The optimization is subject to box constraints on input increments, inputs, and outputs, i.e.

$$\begin{cases} \Delta U_{\min} \le \Delta U \le \Delta U_{\max}, \\ U_{\min} \le U \le U_{\max}, \\ X_{\min} \le X \le X_{\max}. \end{cases} \tag{68}$$

This formulation establishes a convex QP that can be solved online at each sampling instant to obtain the first control increment, after which the procedure is repeated in a receding-horizon approach.

In model predictive control, the choice of the reference state trajectory x_{ref} directly affects the closed-loop transient response, stability margin, and overall robustness. A widely used formulation that balances asymptotic stability and robustness specifies a first-order exponential decay,

$$x_{\text{ref}}(k+i) = x_{\text{des}}(k+i) - \rho^i\left[x_{\text{des}}(k) - x(k)\right], \qquad 0 < \rho < 1, \tag{69}$$

which gradually eliminates the instantaneous tracking error while steering the system toward the desired trajectory x_{des}. The exponential factor $\rho \in (0,1)$ governs how quickly the reference trajectory compensates the instantaneous attitude error along the prediction horizon.

For attitude stabilization, the steady-state objective is

$$\varphi_{\text{des}} = \gamma_{\text{des}} = \dot{\varphi}_{\text{des}} = \dot{\gamma}_{\text{des}} = 0, \tag{70}$$

such that both body angles and their rates vanish, ensuring static equilibrium. Substituting (70) into (69) forces the angular-velocity reference to decay from the currently measured body angular velocity to zero while retaining the same sign. The optimizer therefore keeps damping the speed monotonically rather than injecting an opposite-direction velocity that could cancel the attitude error more aggressively, and the closed-loop reorientation becomes sluggish. To restore satisfactory transient performance without sacrificing asymptotic stability, the angular-velocity reference is designed independently of the attitude reference.

Let the angle and the angular-velocity vectors be denoted by $q = [\varphi, \gamma]$ and $\omega = [\dot{\varphi}, \dot{\gamma}]$, respectively. The adaptive reference for angular velocity is defined as

$$\omega_{\text{ref}}(k+i) = -K_\omega(k) q(k) e^{-\sigma i}, \qquad \sigma > 0, \tag{71}$$

where $K_\omega(k)$ is the adaptive gain matrix synthesized via two logistic (sigmoid) functions, and the decay parameter $\sigma > 0$ governs the exponential decay of the velocity reference.

Introduce the minimum gains $k_{\varphi,\min}, k_{\gamma,\min}$, the maximum gains $k_{\varphi,\max}, k_{\gamma,\max}$, the inflection points φ_0, γ_0 and the slope coefficients c_φ, c_γ, such that $K_\omega(k)$ is calculated by

$$\begin{aligned} K_\omega(k) &= \operatorname{diag}\left\{ k_\varphi^{\text{adapt}}(\varphi(k)), k_\gamma^{\text{adapt}}(\gamma(k)) \right\}, \\ k_\varphi^{\text{adapt}}(\varphi) &= k_{\varphi,\min} + (k_{\varphi,\max} - k_{\varphi,\min}) \frac{1}{1 + e^{-c_\varphi(|\varphi| - \varphi_0)}}, \\ k_\gamma^{\text{adapt}}(\gamma) &= k_{\gamma,\min} + (k_{\gamma,\max} - k_{\gamma,\min}) \frac{1}{1 + e^{-c_\gamma(|\gamma| - \gamma_0)}}. \end{aligned} \tag{72}$$

In this design, $K_\omega(k)$ maintains a low gain in the small-angle regime to suppress sensor noise amplification. When the attitude angle exceeds the predefined inflection point, the gain increases substantially to improve responsiveness under large disturbances. With this adaptive scaling mechanism, expression (72) ensures that the reference angular velocity is oriented in the opposite direction to the instantaneous attitude deviation, exhibits exponential decay along the prediction horizon, and adjusts its magnitude according to the current attitude amplitude. This formulation supports both rapid transient behavior and asymptotic convergence.

Consequently, the four-dimensional reference state is assembled as

$$x_{\text{ref}}(k+i) = \begin{bmatrix} q_{\text{ref}}(k+i) \\ \omega_{\text{ref}}(k+i) \end{bmatrix} = \begin{bmatrix} q_{\text{des}}(k+i) - \rho^i [q_{\text{des}}(k) - q(k)] \\ -K_\omega(k) q(k) e^{-\sigma i} \end{bmatrix}, \tag{73}$$

with $0 < \rho < 1, \sigma > 0$. The parameters ρ and σ govern the global convergence and damping characteristics, while the sigmoid parameters allow for adaptive adjustment of control aggressiveness. This construction provides an attitude reference that guarantees asymptotic convergence while the independently shaped angular-velocity reference supplies the negative impulse required for rapid reorientation.

Remark 1. From the definitions in (69), (71), and (72), the impact of the AMPC parameters on robustness and transient behavior is summarized as follows:

- The exponential factor $\rho \in (0, 1)$ determines the rate at which the reference trajectory compensates the instantaneous attitude error along the prediction horizon. Smaller values of ρ lead to faster convergence toward the desired attitude but may impose higher demands on the control input, resulting in degraded robustness. Larger values of ρ yield smoother transitions with slower error decay, enhancing stability under model uncertainties and actuation limits.
- The decay parameter $\sigma > 0$ shapes the exponential attenuation of the velocity reference. Larger values of σ cause stronger damping and faster stabilization, while smaller values of σ preserve agility but reduce robustness against disturbances.
- The minimum gains $k_{\varphi,\min}$ and $k_{\gamma,\min}$ ensure baseline robustness in the near-zero attitude region. The maximum gains $k_{\varphi,\max}$ and $k_{\gamma,\max}$ limit corrective effort to avoid actuator saturation. The attitude thresholds φ_0 and γ_0 designate the inflection points that separate the small-angle regime from larger angular deviations that call for more aggressive control action. The slope coefficients c_φ and c_γ determine the rate of gain transition around the inflection points. Smaller coefficient values produce gradual gain changes, which contribute to the attenuation of sensor noise. In contrast, larger coefficients result in steeper transitions that enable faster gain amplification in response to abrupt attitude deviations, enhancing the disturbance rejection performance.

The tuning procedure of the AMPC parameters is given as follows. The admissible ranges are set from stability, constraint feasibility, and real time requirements, guided by the analytical roles of ρ, σ, and the sigmoid parameters summarized above. Within these ranges, a coarse sweep varying one parameter at a time is run to obtain a safe baseline. Then, a short local search around the baseline is performed to balance tracking and robustness, evaluating settling behavior, overshoot, disturbance rejection, total control variation, constraint satisfaction, and solver time under representative disturbances. Final values are reported in Table 3 and the same set is used in all experiments.

6. Distributed control framework for the wheeled hexapod rescue device

Based on the control methods presented above, the distributed control framework for the wheeled hexapod rescue device is illustrated in Fig. 5. The operator remotely supervises the rescue device through the teleoperation console using real-time video streams from the onboard perception system and issues locomotion commands to both the mechanical arm and the chassis via joysticks.

For the mechanical arm, the desired end-effector pose commanded from the console is transformed into joint motor angles and electric cylinder position commands via inverse kinematics, constrained optimization, and electric cylinder length mapping, thereby enabling coordinated arm movement. In practical operations, the motion of the wheeled hexapod chassis and the manipulation of the arm are executed sequentially. Specifically, the chassis navigates to a position near the target object and stabilizes its body attitude without any motion of the mechanical arm. Subsequently, the mechanical arm is commanded to perform grasping, lifting, or relocating tasks, while the chassis remains stationary. This decoupled procedure ensures that the arm operates on a stable base, avoiding mutual interference between locomotion and manipulation, which guarantees the stability of the robotic system.

For chassis locomotion, the top-level attitude stabilization controller first processes the desired velocity, orientation, and attitude issued from the console. It then computes the reference foot-end positions x_r and the reference electric cylinder forces f_r for each wheel-leg. These are then passed to the bottom-level force-position hybrid controller, which outputs the final position commands

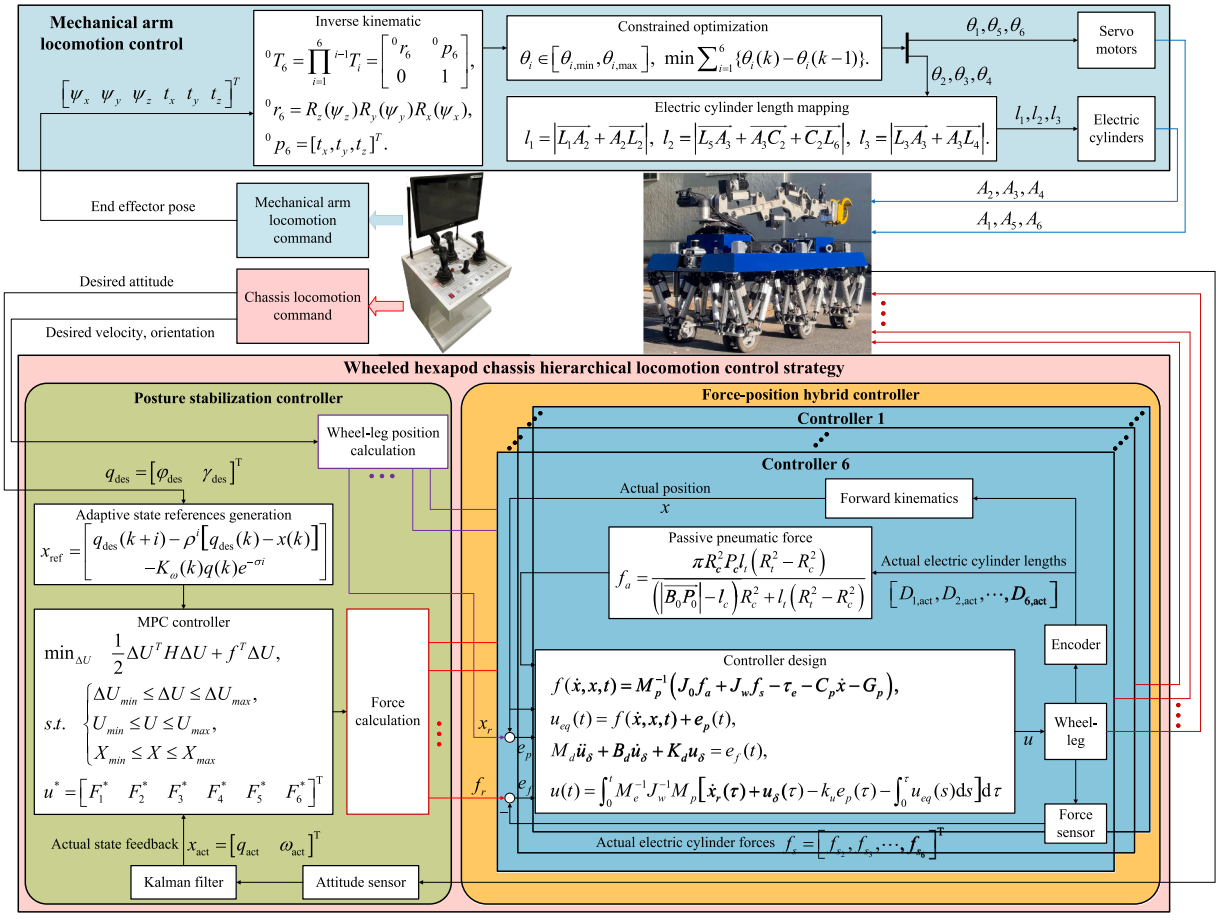


Fig. 5. Distributed control framework for the wheeled hexapod rescue device. Mechanical arm locomotion control converts desired end-effector poses into joint angles and cylinder positions via inverse kinematics and optimization. Hierarchical chassis locomotion control employs AMPC for attitude stabilization and a force-position hybrid controller to track reference foot-end positions and cylinder forces.

for each electric cylinder. To resolve the potential timing conflicts between the top-level AMPC and the bottom-level force-position control, the top-level AMPC runs at 100 Hz, while each bottom-level controller operates at 500 Hz. Communication between layers is implemented via UDP over a 100 Mb fiber-optic Ethernet link, ensuring reliable transmission of control commands at 100 Hz. The bottom-level controllers adopt a zero-order-hold (ZOH) scheme, maintaining the most recently received command until the next update. In addition, the effective road excitation frequency f_{road} is estimated as

$$f_{\text{road}} = \frac{v}{\lambda}. \tag{74}$$

where λ and v represent the characteristic terrain wavelength and robot locomotion velocity, respectively. Based on the result of [36], the minimal dominant wavelength of road surface irregularities is selected as $\lambda_{\min} = 0.5$ m. Since the maximum wheeled locomotion velocity of the proposed robotic system is $v_{\max} = 15$ km/h ≈ 4.17 m/s, the maximum road excitation frequency is 8.33 Hz. Consequently, the selected 100 Hz control frequency is above twice this value, which satisfies the Nyquist sampling theorem. This ensures that the AMPC can effectively capture and compensate for attitude variations induced by realistic ground irregularities in real time, with sufficient robustness margin. In this way, chassis motion is controlled in a distributed hierarchical architecture.

7. Experiments

This section presents experimental studies conducted to evaluate the proposed rescue device and the distributed control framework. The energy efficiency and tracking accuracy of the wheel-leg with pneumatic transmission, the terrain adaptation performance of the hierarchical locomotion control strategy, and the terrain traversal performance under representative scenarios are examined. Furthermore, teleoperation tests, overall performance comparison, and field applications are conducted to validate the effectiveness and reliability of the whole system.

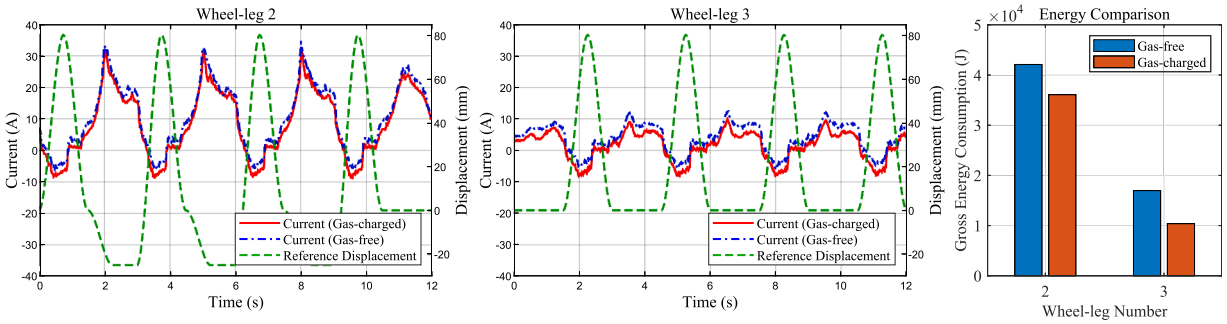


Fig. 6. Energy efficiency comparison of the pneumatic transmission. (Left) Reference foot-end displacement (green) and total cylinder current under gas-charged (red) and gas-free (blue) conditions for Wheel-leg 2. (Center) Same metrics for Wheel-leg 3. (Right) Gross energy consumption comparison under both conditions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Time-delay-compensated displacement tracking errors for Wheel-leg 2 and Wheel-leg 3 under gas-charged and gas-free conditions.

Condition	Max $\|e(t)\|$ (mm)		RMSE (mm)	
	Wheel-leg 2	Wheel-leg 3	Wheel-leg 2	Wheel-leg 3
Gas-charged	5.01	3.83	2.03	1.87
Gas-free	4.93	4.16	2.07	1.92

7.1. Energy efficiency and tracking performance

Pneumatic transmission is designed to counteract the gravitational load, reducing energy consumption and improving endurance and payload capacity. To verify its energy-saving effect, in-place stepping tests were conducted under two conditions: gas-charged and gas-free. The results are shown in Fig. 6. Owing to the robot's structural symmetry, only the reference displacement and current profiles of a central leg (Wheel-leg 2) and a corner leg (Wheel-leg 3) are presented to represent all wheel-legs. The left and center panels display each wheel-leg's reference foot-end displacement (green curve) and total cylinder current under gas-charged (red) and gas-free (blue) conditions. As evident in the plots, the current under gas-charged conditions is generally lower than under gas-free conditions, qualitatively demonstrating the energy-saving characteristic of the pneumatic transmission.

For a quantitative analysis, the gross energy consumption under both conditions was computed according to

$$E = \int_0^T U \max\{I(t), 0\} \mathrm{d}t, \tag{75}$$

where T, U and I denote the test duration, bus voltage and the total current of cylinders, respectively. Since regenerative currents ($I < 0$) are dissipated as heat in braking resistors instead of being returned to the power bus, only the positive currents ($I > 0$) represent net energy drawn from the supply. The gross energy comparison is shown in the right panel of Fig. 6. Energy consumption under gas-charged conditions is lower than under gas-free conditions, and the energy-saving rates of the pneumatic transmissions for Wheel-leg 2 and Wheel-leg 3, calculated by

$$\eta = \frac{E_{\text{Gas-free}} - E_{\text{Gas-charged}}}{E_{\text{Gas-free}}} \times 100\%, \tag{76}$$

are 14.28% and 38.79%, respectively.

In addition, to explicitly quantify the displacement tracking accuracy and stability of the proposed force-position hybrid control strategy, the displacement tracking results of Wheel-leg 2 and Wheel-leg 3 obtained from the same in-place stepping experiment are presented in Fig. 7.

Specifically, the reference and measured foot-end displacement trajectories under the gas-charged and gas-free conditions are shown in Fig. 7a and b, respectively. As illustrated by the tracking curves, the measured displacement exhibits a small phase lag (approximately $\tau \approx 0.13$ s) with respect to the reference trajectory due to actuator dynamics and closed-loop response. The phase lag τ is estimated by aligning the rising edges of the measured and reference displacement trajectories. In this case, the tracking error is dominated by the phase lag component. Therefore, to evaluate the tracking accuracy in terms of displacement magnitude, we additionally compute a time-delay-compensated error $e(t) = x(t) - x_r(t - \tau)$, which is reported in Fig. 7c. The corresponding quantitative error metrics, namely the maximum absolute error and the root mean squared error (RMSE), are summarized in Table 2.

As summarized in Table 2, the time-delay-compensated tracking errors remain small and exhibit high consistency across gas-charged and gas-free conditions. These results indicate that the proposed force-position hybrid control strategy achieves satisfactory

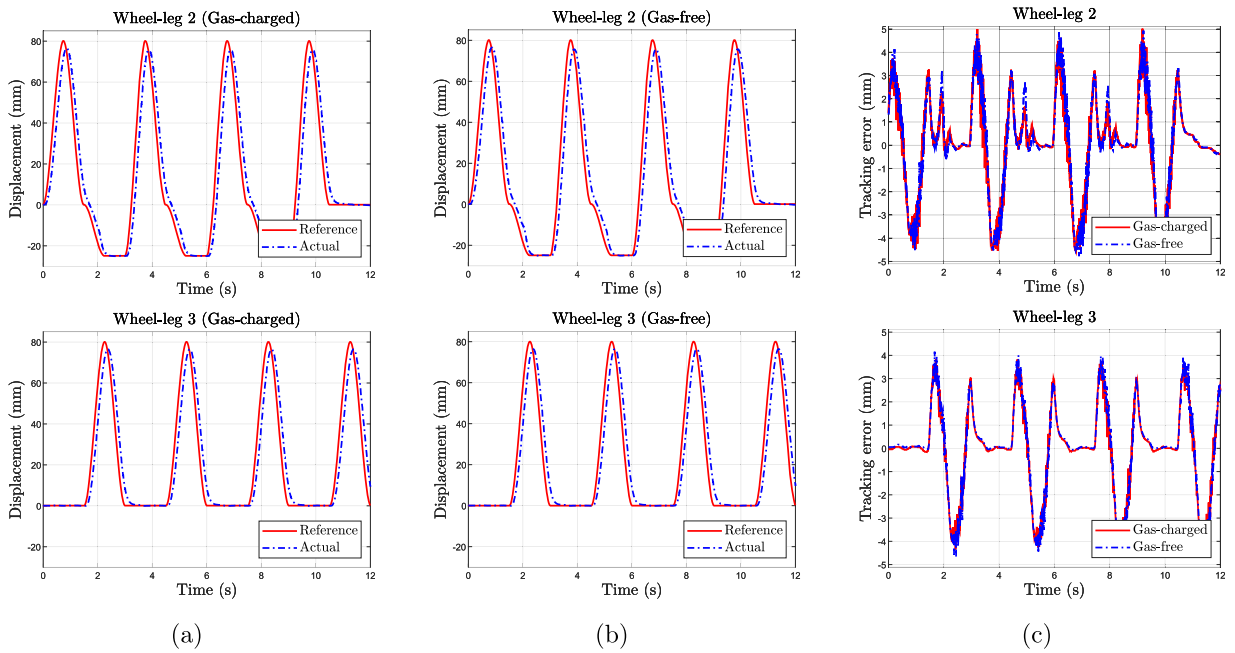


Fig. 7. Displacement tracking performance. (a) Gas-charged tracking results. (b) Gas-free tracking results. (c) Time-delay-compensated tracking errors.

displacement tracking accuracy with bounded errors, and that the pneumatic assistance does not compromise the tracking accuracy or the stability of the closed-loop tracking performance.

7.2. Terrain adaptation performance

To validate the effectiveness and reliability of the hierarchical locomotion control strategy, comparative tests were conducted on a continuous and smoothly varying sloped terrain. In all experiments, the AMPC controller was executed with a sampling period $T = 0.01$ s, and both the prediction horizon p and the control horizon q were set to 20. To justify the selection of prediction and control horizons in the AMPC, numerical tests were conducted to systematically evaluate the control input convergence and computational cost under different horizons. For implementation simplicity, both prediction and control horizons were set to the same length in all tests to avoid the horizon mismatch. The results are summarized in Fig. 8.

Fig. 8a illustrates that the optimized control inputs converge as the horizon increases, with negligible changes beyond 15 steps. Due to the structural symmetry of the robot, the attitude-regulation forces exhibit symmetric patterns with respect to both the longitudinal and lateral axes. Moreover, the AMPC solution is a coupled optimization vector, such that all force components converge simultaneously as the prediction horizon increases. Therefore, forces F_1 to F_3 are plotted for clarity of presentation, which sufficiently capture the convergence trend of the solution. Fig. 8b reports that the corresponding solve time remains below 3×10^{-3} s from 0 to 30 steps, which fulfills the real-time requirement for the loop frequency of 100 Hz. Additionally, the solve time exhibits a positive correlation with the horizon.

Although the optimized input essentially saturates for horizons of 15 steps, we chose the prediction and control horizon of 20 steps in implementation. A longer horizon mitigates finite-horizon truncation and improves tolerance to model–plant mismatch, contact-timing variability, and sensing delays by allowing the optimizer to anticipate and compensate future deviations earlier. The increasing prediction horizon further strengthens the smoothness of control actions, which benefits actuation safety and stability on rough terrain. Moreover, this conservative buffer enhances uniformity of performance across operating conditions (e.g., different initial attitudes and terrain contacts), while the extra computational cost is negligible, i.e., the measured solve time at 20 steps remains below 2 ms. It is noted that the solve time rises above 2 ms from 21 steps onward. In our implementation, the AMPC solver shares the same control cycle with other essential processes such as sensor data acquisition, filtering, and service handling. Allowing the AMPC solve time to exceed 2 ms would reduce the computational margin available for these concurrent tasks, causing degraded real-time responsiveness. Limiting the horizon to 20 steps ensures that the solve time remains below 10 ms while preserving sufficient margin for other processes.

The AMPC controller parameters are listed in Table 3, and the key structural parameters relevant to control performance are given in Table 4. Figs. 9 and 10 illustrate the motion behavior and pitch angle variations of the robot without and with the proposed attitude stabilization control, respectively.

As shown in Fig. 9, in the absence of stabilization control, the robot exhibits large pitch fluctuations up to 9° during both uphill and downhill motion. Additionally, foot-end detachment from the ground is observed, as highlighted by the red circles in the figure.

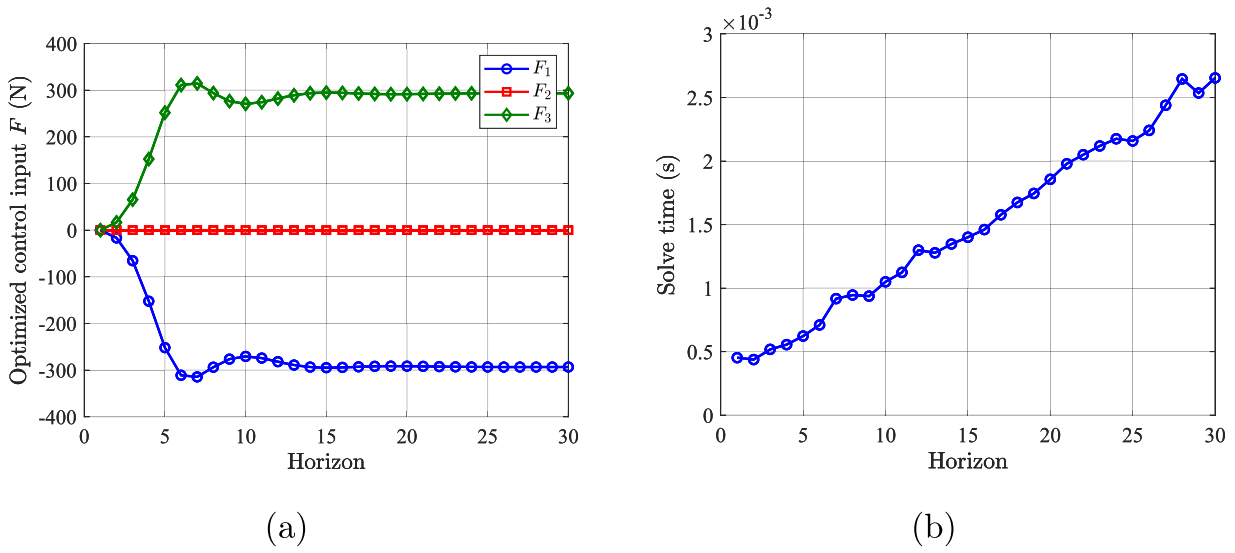


Fig. 8. Numerical tests of the AMPC controller under different prediction and control horizons. (a) The convergence of the optimized control inputs. (b) The corresponding solve time.

Table 3
AMPC controller settings.

Parameters	Values	Parameters	Values	Parameters	Values	Parameters	Values
T	0.01 s	p, q	20, 20	ρ	0.92	σ	0.50
$k_{\varphi,\min}$	0.50	$k_{\varphi,\max}$	1.50	$k_{\gamma,\min}$	1.00	$k_{\gamma,\max}$	2.00
c_φ	10.00	c_γ	10.00	φ_0	0.80	γ_0	2.00

Table 4
Structural parameters of the rescue device.

Parameters	Values	Parameters	Values
Body length	2.90 m	Body width	1.50 m
Initial body height	1.30 m	Two-stage cylinder stroke	600 mm
Wheel-leg mass	123.5 kg	Body mass	721.6 kg
Chassis mass	1462.6 kg	Mechanical arm mass	317.4 kg

In contrast, Fig. 10 shows that with the application of the proposed hierarchical strategy, combining force-position hybrid control for the wheel-legs and attitude stabilization, the robot actively adjusts the length of each leg in response to terrain undulations, actuator forces, and body attitude. This ensures reliable foot-end ground contact and maintains the pitch angle within ±4°, thereby enhancing overall motion stability.

7.3. Terrain traversal performance

To validate the terrain traversal performance of the proposed device, obstacle-crossing experiments were conducted. On roads affected by landslides or collapses, rockfall debris may lead to limited bypass space and obstacles whose width exceeds the nominal track width of the robot. In such cases, the robot can cross the obstacle through track-width extension. As shown in Fig. 11, an obstacle-crossing experiment was conducted to simulate this terrain condition. The obstacle width was approximately 56 cm. By lateral stepping based on a tripod gait, the robot increased its inner-side track width to approximately 136.4 cm and traversed the obstacle directly. Throughout the maneuver, three supporting legs remained in contact with the ground, ensuring body stability during obstacle crossing.

On rockfall-affected roads, obstacle height may exceed the safe underbody clearance of the robot in its nominal configuration. The safe underbody clearance is smaller than the nominal body height because the battery and the engine are mounted beneath the body. In such terrain, the robot can traverse the obstacle through body elevation. As shown in Fig. 12, the robot crossed an obstacle approximately 85 cm high and then lowered its body to resume forward motion. Benefiting from the long stroke of the two-stage leg cylinders, the robot can significantly elevate the body to increase the safe clearance and pass over such obstacles. The adjustable body height ranges from 130 cm to 180 cm, with the corresponding safe clearance ranging from 70 cm to 120 cm.

When the robot encounters trenches or elongated protruding obstacles, track-width extension or body elevation alone may be insufficient. In such cases, the robot can traverse the terrain by sequential leg adjustment while keeping the projection of the CoM

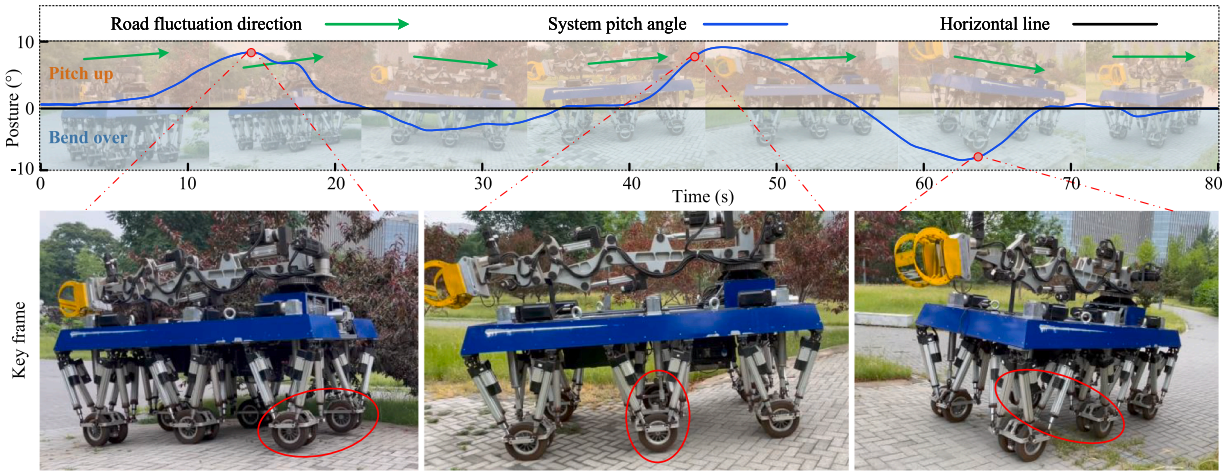


Fig. 9. Without the hierarchical locomotion control strategy, pitch angle fluctuates up to 9° along with foot-end detachment from the ground, as indicated by red circles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

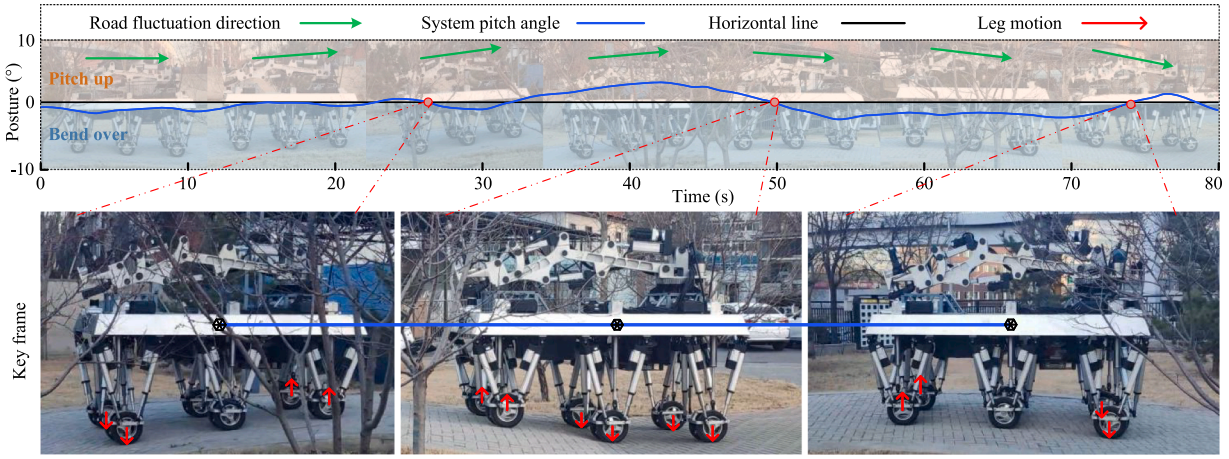


Fig. 10. With the hierarchical locomotion control strategy, real-time leg-length adjustment is enabled based on terrain feedback, ensuring reliable foot-end contact and maintaining the pitch angle within ±4°.

within the support polygon. The trench-crossing experiment is shown in Fig. 13a, where tape was used to simulate trench edges with a width of approximately 100 cm. During traversal, legs 1 and 6 were first suspended above the trench, while legs 2 and 5 were swung forward and legs 3 and 4 were swung backward to reconfigure the support polygon. The same principle was applied when the other leg pairs were suspended. Fig. 13b shows a similar experiment for a one-meter-wide elongated protruding obstacle formed by stacked bricks. The robot used the same sequential leg-adjustment strategy and increased the body clearance to traverse the obstacle while maintaining a valid support polygon.

To further evaluate the terrain traversal capability on a steep slope, an additional climbing experiment was conducted on a rigid inclined surface with a slope angle of 30°, as shown in Fig. 14. The safety ropes were used only as protective restraints to prevent accidental damage during the climbing test and did not provide additional traction or load support to the robot. The robot completed the climbing process in a stable manner, indicating that the proposed device can traverse a 30° slope under the tested conditions.

7.4. Field application

The wheeled hexapod robotic system can be remotely operated via a teleoperation console connected through a wireless network, enabling execution of complex operational tasks. Fig. 15 illustrates a simulated teleoperation scenario, where the operator remotely navigates the robot to the task site using joysticks, manipulates the mechanical arm based on real-time visual feedback from the onboard perception system, and subsequently commands the robot to retreat from the operation area.

In addition to the teleoperation navigation scenario, a grasping experiment was conducted to evaluate the stability of the robotic system during manipulator operations. As shown in the first frame of Fig. 16, the robot was teleoperated to approach a target object

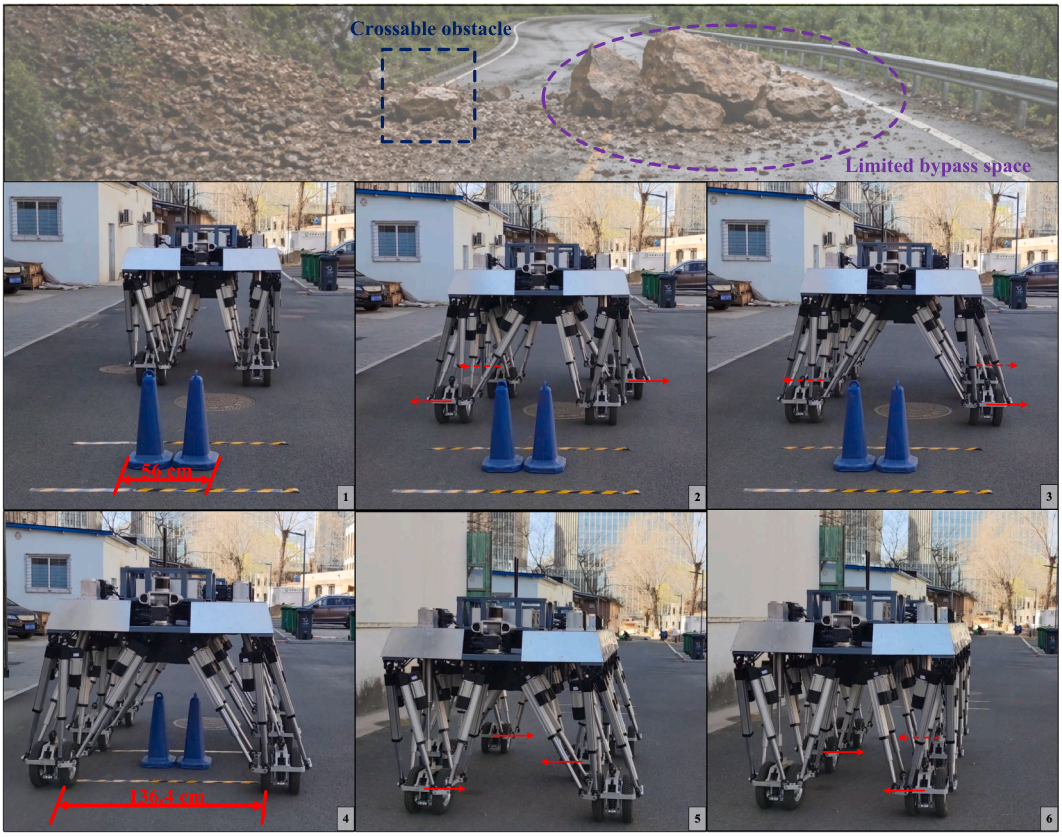


Fig. 11. Obstacle-crossing via track-width extension in a simulated rockfall scenario with limited bypass space and a wide obstacle.

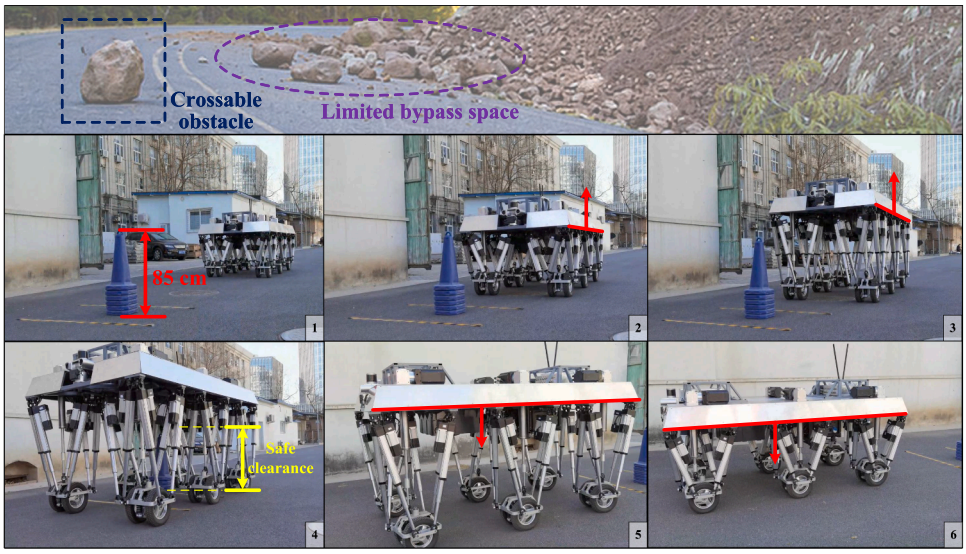
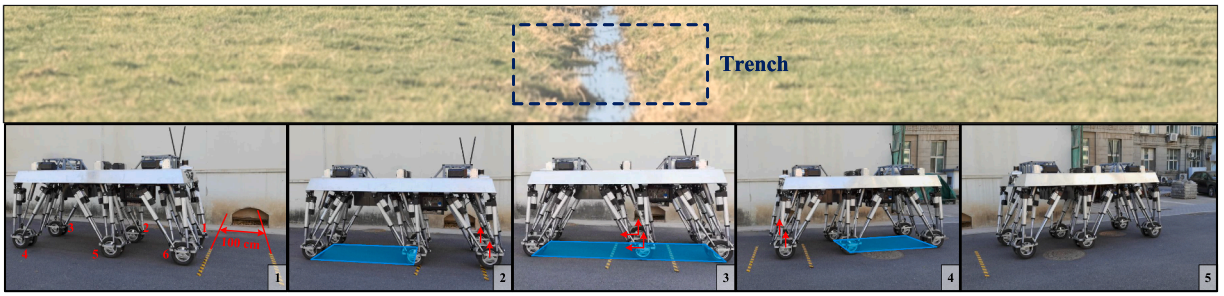
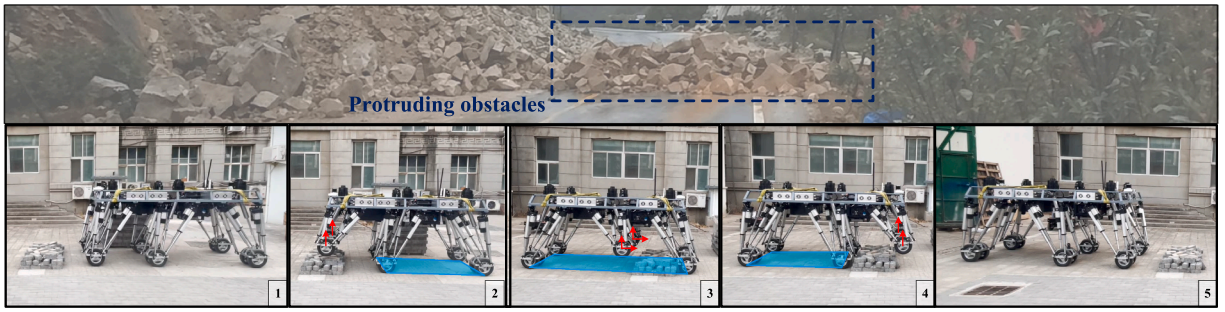


Fig. 12. Obstacle-crossing via body elevation in a simulated rockfall scenario with limited bypass space and a high obstacle.



(a)



(b)

Fig. 13. Obstacle-crossing via sequential leg adjustment. (a) Crossing a simulated trench. (b) Crossing a simulated elongated protruding obstacle.

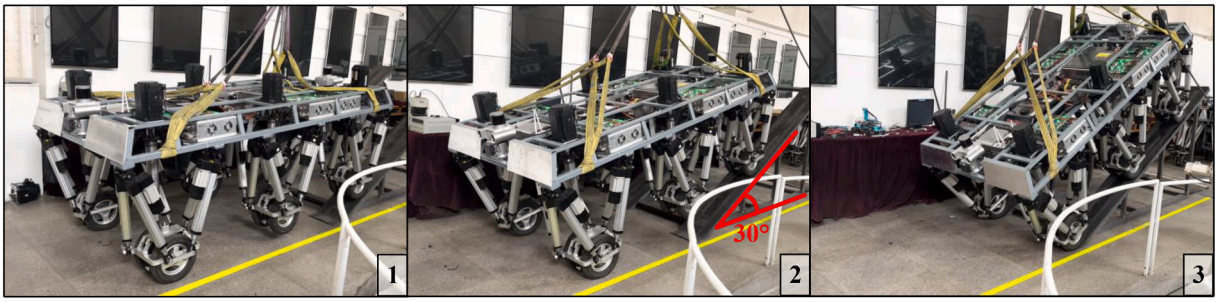


Fig. 14. Slope-climbing experiment on a rigid 30° inclined surface.

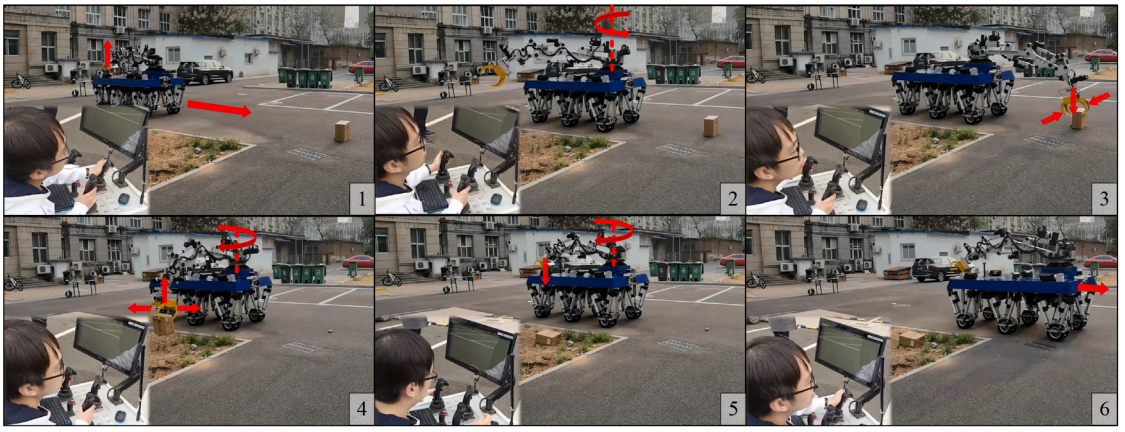


Fig. 15. Teleoperation of the robot via a remote teleoperation console through a wireless network.

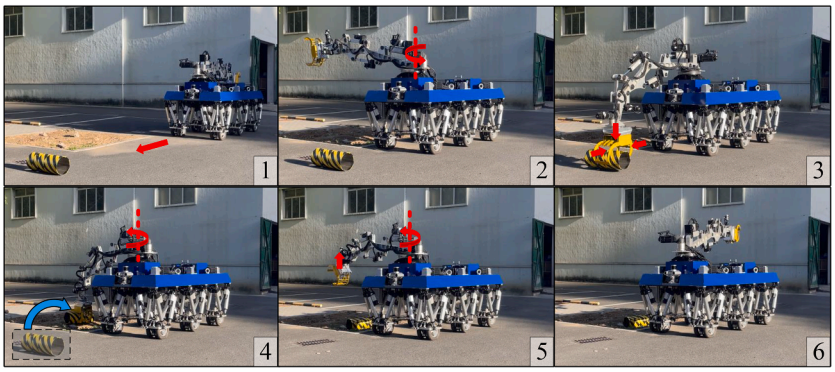


Fig. 16. Grasping experiment using the mechanical arm.

Table 5
Comparison of the proposed rescue device with BITNAZA, ASTERISK H, and ATHLETE.

Performance indicator	ASTERISK H	ATHLETE	BITNAZA	Proposed device
Total mass (kg)	3.4	850	378	1780
Body length (m)	0.815	2.75	1.8	2.9
Body width (m)	0.537	2.75	1.8	1.5
Wheel-leg mechanism	Serial	Serial	Parallel	Parallel
With pneumatic transmission	No	No	No	Yes
Payload capacity (kg)	–	1100	350	1200
Endurance (h)	–	–	3	4 (Battery) or 12 (Hybrid)
Track width range (m)	–	–	0.66–1.26	0.70–1.70
Track width span (m)	–	–	0.60	1.00
Body height range (m)	–	–	1.17–1.57	1.20–1.80
Body height span (m)	–	–	0.40	0.60
Maximum traversable slope (°)	–	–	27	30
Manipulation	No	No	No	Yes

and stabilized at the task site. The remaining frames indicate that the mechanical arm was commanded to execute a sequence of grasping and lifting motions, followed by placing the object to a different position. Throughout the entire process, the wheel–legged platform maintained a stable body attitude without significant disturbance, confirming that locomotion and manipulation can be performed in a sequential and reliable manner.

A comparison between the proposed wheeled hexapod rescue device and the wheeled hexapod robots BITNAZA [34], ASTERISK H, and ATHLETE [28] is presented in Table 5. For ASTERISK H, ATHLETE, and BITNAZA, the values in Table 5 are taken from the corresponding references. For the proposed device, all values are measured on the prototype. As summarized in Table 5, ASTERISK H and ATHLETE employ serial mechanisms, while BITNAZA and the proposed device adopt Stewart-structured parallel mechanisms. This parallel architecture distributes ground reaction forces and reduces joint torque demand for a given foot force, which is beneficial for payload capacity. The pneumatic transmission provides passive forces to support the weight of the platform as well as external payloads, contributing to the achieved payload capacity of 1200 kg while helping reduce energy consumption. The hybrid power system provides an endurance of 4 h in battery-only mode and 12 h in hybrid-power mode. In addition, the wide adjustment ranges of track width and body height, together with their corresponding adjustment spans, enlarge the geometric workspace for obstacle crossing and posture adaptation in unstructured terrains. Compared with BITNAZA, the maximum traversable slope angle increases from 27° to 30°, indicating improved traversability on steep slopes. Finally, the mechanical arm extends the platform functionality from locomotion to object handling in complex rescue scenarios. Therefore, the proposed device integrates terrain adaptability, sufficient payload capacity, low energy demand for sustained deployment, and stable body attitude for manipulation and transport, which is consistent with the practical requirements of rescue missions.

To further assess system stability and terrain adaptability, field trials were conducted on rough, sloped surfaces with scattered small gravel and local surface unevenness. As shown in Fig. 17, the system maintained a stable body attitude and consistent foot-end contact during traversal of the uneven sloped terrain, demonstrating the effectiveness of the proposed control algorithms. Additional field experiments were performed at a firefighting and disaster-relief drill site. As depicted in Fig. 18, the rescue device exhibited stable locomotion over unstructured outdoor environments such as sand and loose soil. The mechanical arm successfully executed manipulation tasks, validating the safety, reliability, and applicability of the proposed system for practical rescue operations.

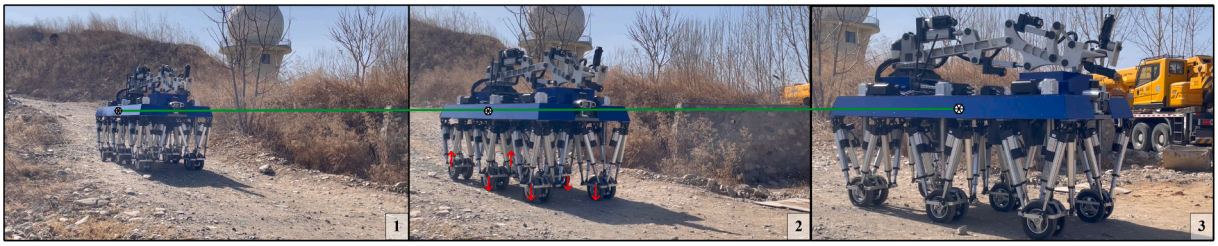


Fig. 17. Field test of the rescue device on a rough sloped terrain with scattered small gravel and local surface unevenness. The device maintains reliable ground contact and a stable body attitude while traversing the uneven slope, demonstrating the effectiveness of the proposed control strategy.

Fig. 18. Field application in firefighting and disaster-relief scenarios. The rescue device demonstrates stable locomotion over unstructured terrain such as sand and soil.

8. Discussion

This study presents a comprehensive design and control framework for a wheeled hexapod system capable of traversing complex terrains and performing manipulation tasks in disaster scenarios. The integration of a Stewart-structured wheel-leg with pneumatic-assisted actuation enhances terrain adaptability and load capacity while improving energy efficiency through gravity compensation. The control framework, composed of the force-position hybrid controller for wheel-legs and AMPC for body attitude stabilization, enables reliable locomotion in challenging conditions, including lateral obstacle crossing, body elevation, and trench traversal, while maintaining body stability and minimizing attitude deviations. The adaptive angular-velocity reference generation within the AMPC formulation further improves transient responsiveness without compromising steady-state accuracy. The inclusion of a mechanical arm, coordinated through inverse kinematics and constrained optimization, extends the operational capabilities of the system. Experimental validation confirms that the pneumatic subsystem contributes to reducing actuator energy consumption. Real-world field trials show that the proposed system can perform manipulation tasks such as lifting and relocating objects in visually and physically complex environments.

In terms of terrain adaptability, the proposed device provides a track width adjustment range of 0.7–1.7 m and a body height adjustment range of 1.2–1.8 m, with a corresponding safe underbody clearance of 0.7–1.2 m. In the experiments, the robot crossed a 0.56 m wide obstacle through track-width extension, a 0.85 m high obstacle through body elevation, a 1.0 m trench and a 1.0 m elongated protruding obstacle through sequential leg adjustment, and climbed a 30° slope. These results indicate that the proposed rescue device and control framework are applicable to representative rescue terrain conditions such as rockfall-affected roads, trenches, elongated protruding obstacles, and rough slopes. Under these conditions, terrain traversal can be achieved through wheel-leg reconfiguration and terrain-adaptive locomotion. However, the applicability of the current design has not been validated for more extreme terrains, such as giant boulder fields, deep soft ground that may submerge the wheel hubs, or steep slopes with loose surfaces. Future work will investigate these conditions through improvements in the mechanical structure, terrain perception, and locomotion control.

CRediT authorship contribution statement

Shangfei Liu: Writing – original draft, Validation, Software, Methodology, Investigation; **Junzheng Wang:** Resources, Project administration, Funding acquisition, Conceptualization; **Dongchen Liu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition; **Huaihang Zheng:** Validation, Software, Methodology, Investigation.

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Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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